08th Sep 2026

1 1.008 **H** Hydrogen | 92 238.03 **U** Uranium | 9 19.00 **F** Fluorine | 16 32.06 **S** Sulfur

수소경제와 에너지저장 시스템

(Hydrogen Economy and Energy Storage Systems)

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Hydrogen **E**nerg**Y** Lab **ON** **E**arth

Assistant Professor

Division of Climate Change

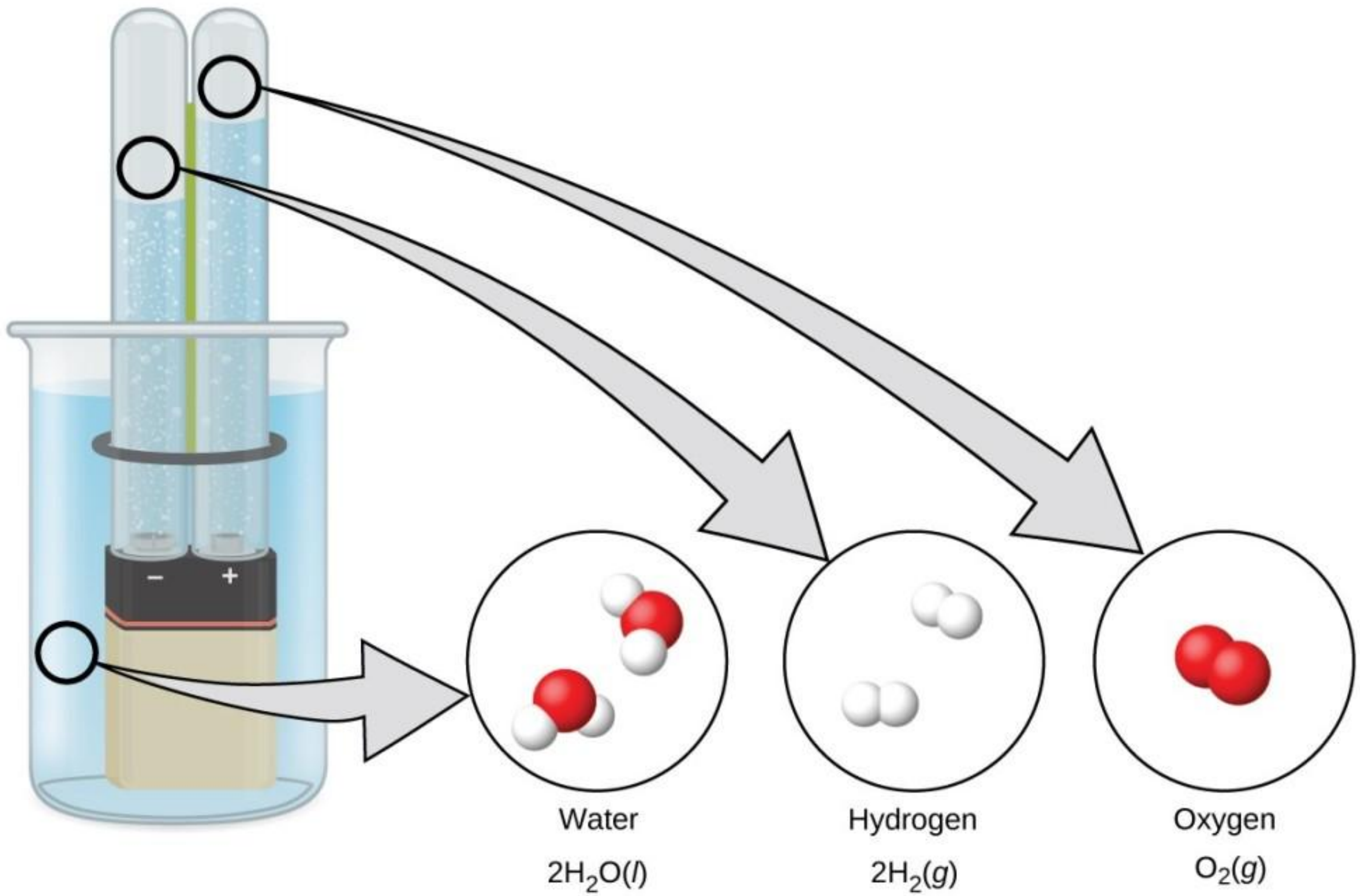
Hankuk University of Foreign Studies (HUFS)

2. Introduction to Hydrogen (2)

Contents

Water Splitting

❑ Electrochemical Reduction of Water

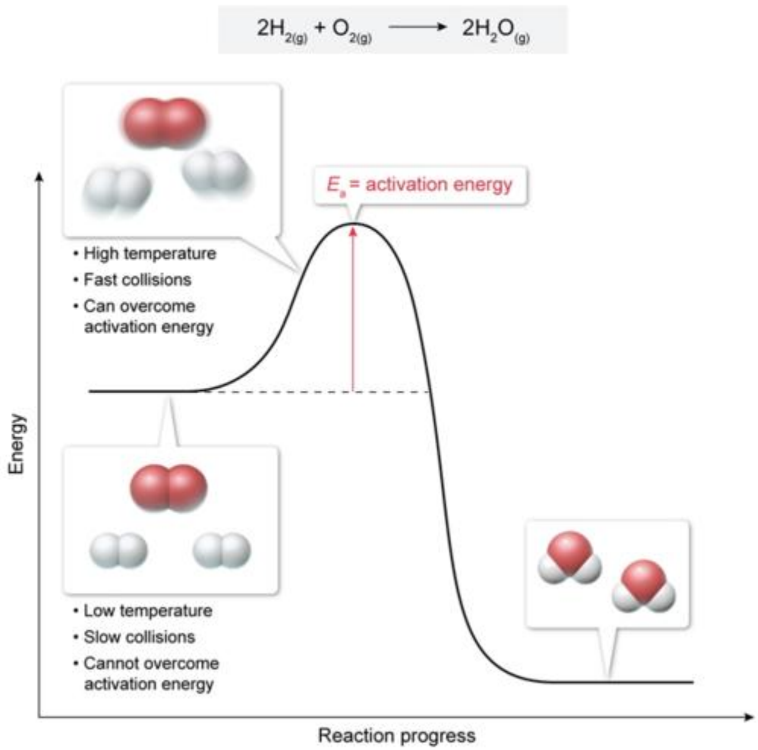


Hydrogen Energy

❑ Oxidation of Hydrogen

Exothermic reaction where hydrogen burns with oxygen to produce water/steam and heat

- **Higher-Heating Value (HHV)**
  $H_2(g) + 0.5O_2(g) \rightarrow H_2O(l)$
  $\Delta H° = -285.8$ kJ mol^{-1} (=~141.8 MJ kg^{-1})
  $\Delta G° = -237.1$ kJ mol^{-1}
- **Lower-Heating Value (LHV)**
  $H_2(g) + 0.5O_2(g) \rightarrow H_2O(g)$
  $\Delta H° = -241.8$ kJ mol^{-1} (=~120.0 MJ kg^{-1})
  $\Delta G° = -228.6$ kJ mol^{-1}



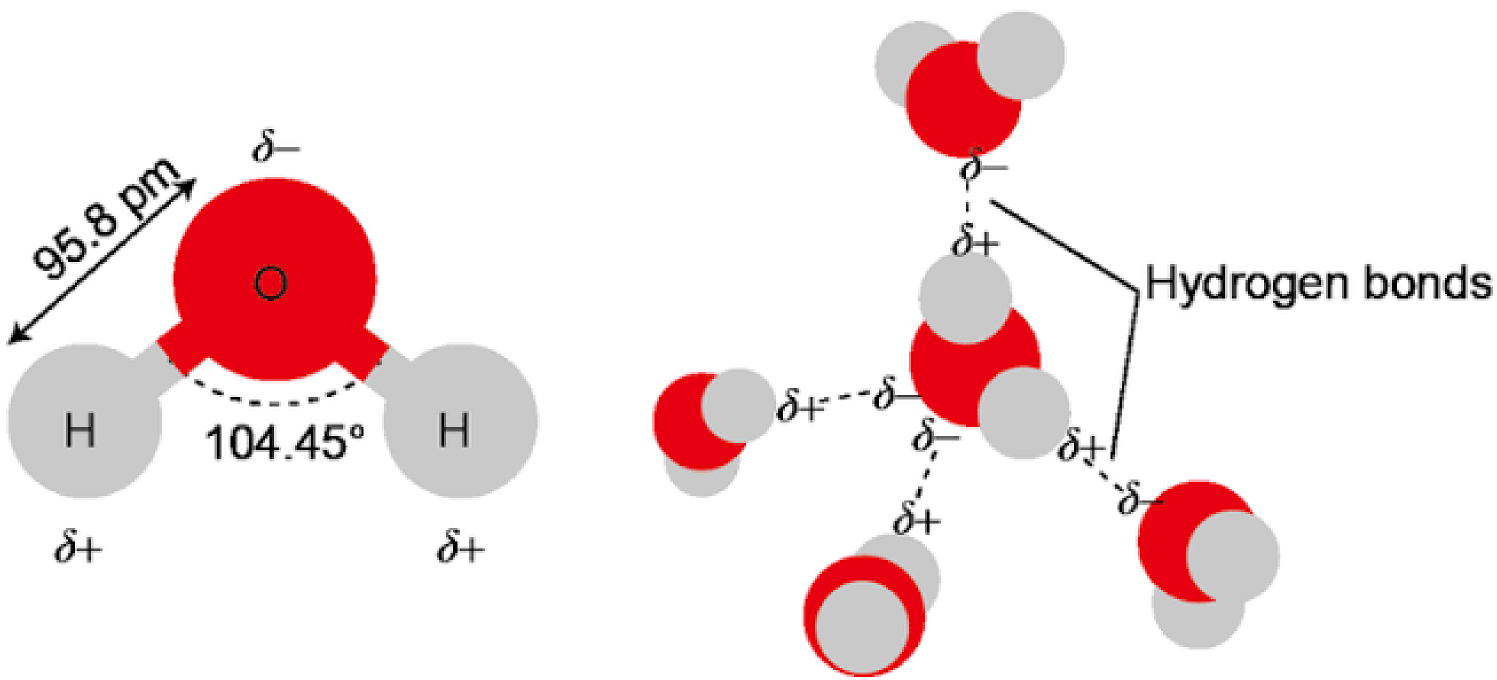
Temp (°C)	ΔH (kJ mol^{-1})	ΔG (kJ mol^{-1})
25	–285.8	–237.1
50	–285.0	–233.1
100	–283.5	–225.2
150	–243.1	–222.8
200	–243.6	–220.4
300	–244.5	–215.4
400	–245.4	–210.2
500	–246.2	–204.9
600	–247.0	–199.5
700	–247.7	–194.1
800	–248.3	–188.5

Water Splitting

❑ Structure of Water

Consists of one oxygen atom joined to two hydrogen atoms by single polar covalent bonds

- Bent structure with a bond angle of 104.5°, caused by two lone pairs on the oxygen atom
- O–H bond distance of 95.8 pm (=0.958 Å = 9.58×10^{-11} m)
- Abnormally high boiling point (100 °C), high surface tension (72.8 mN m^{-1} at 20 °C), and density anomaly upon freezing



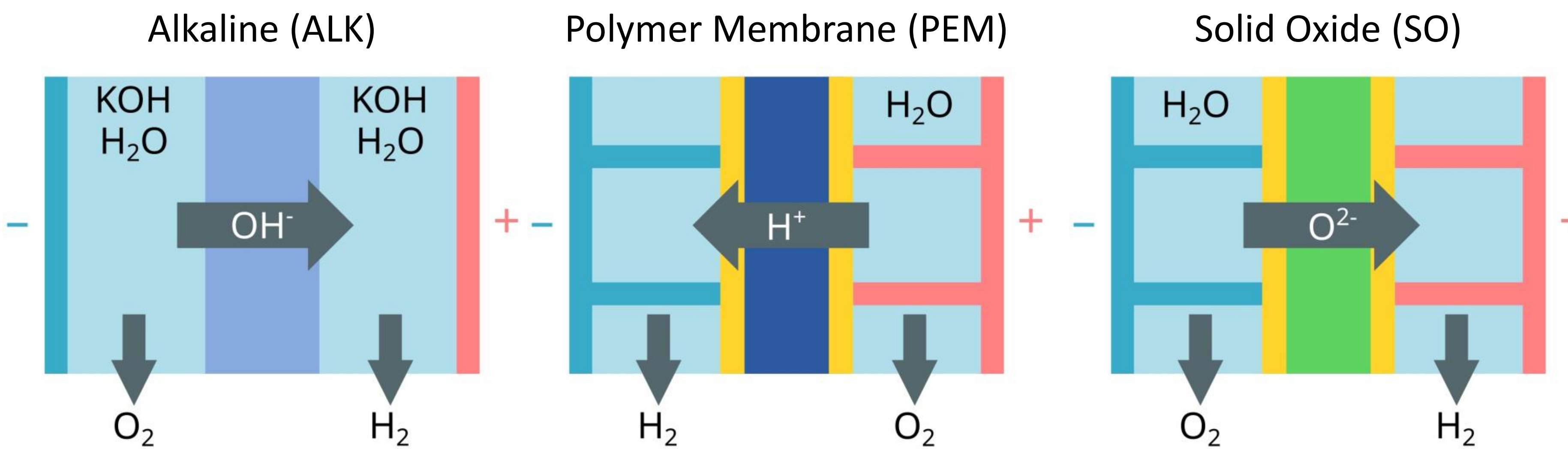
Sci. Sin. Chim., **49**(3) (2019) 536-555

Electrolysis Hydrogen

❑ Water (Steam) Electrolysis

A process of using direct electric current to split water into hydrogen and oxygen

- $H_2O(l \text{ or } g) \rightarrow H_2(g) + 0.5O_2(g)$



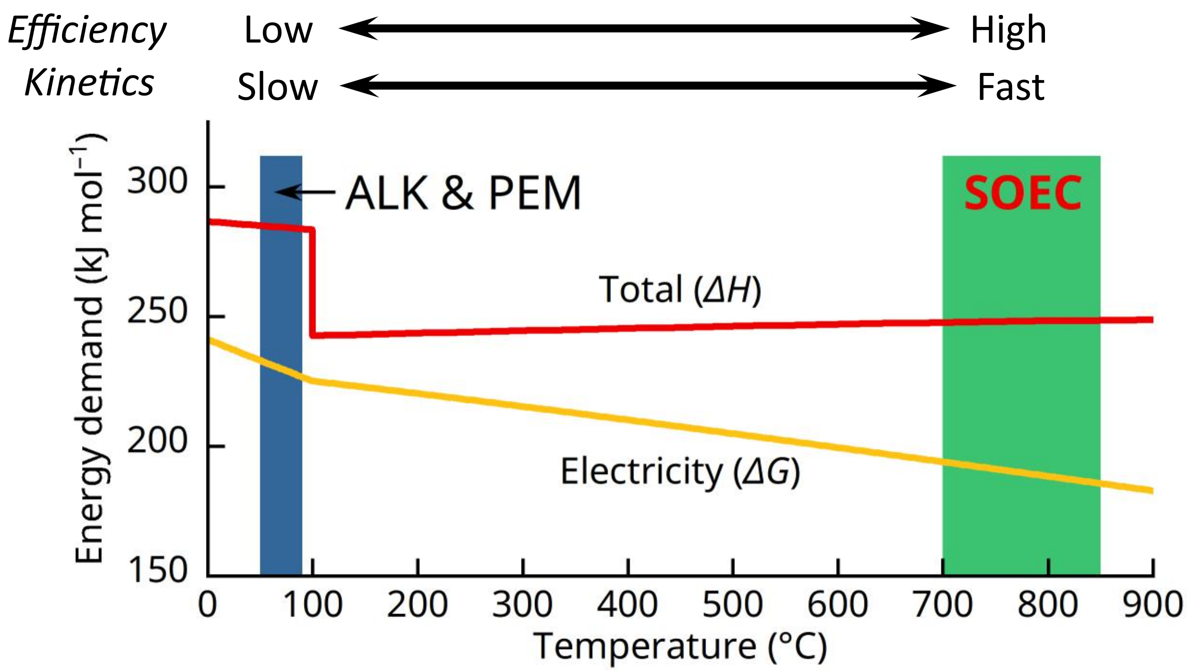
Small prototype				Large prototype		Demonstration		Market uptake		Mature
1						SO		ALK PEM		11

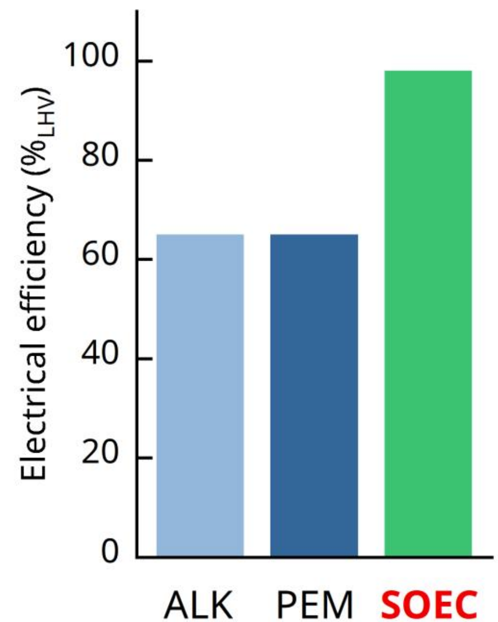
<Source: IEA Electrolyser>

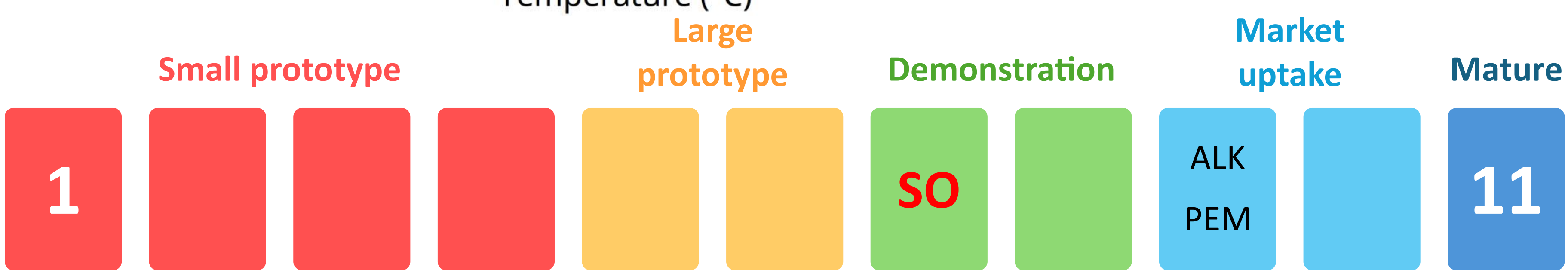
Electrolysis Hydrogen

❑ Water (Steam) Electrolysis

A process of using direct electric current to split water into hydrogen and oxygen







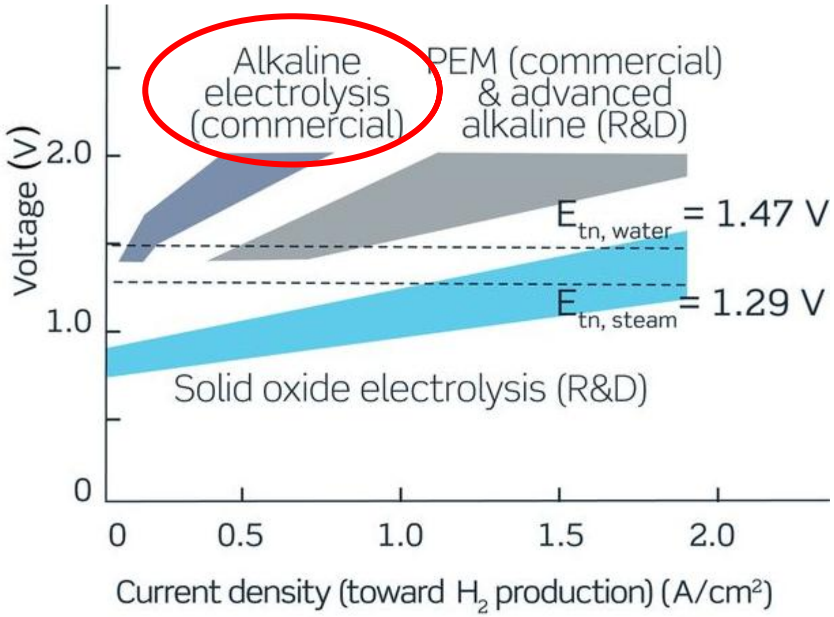
<Source: IEA Electrolyser>

Electrolysis Hydrogen

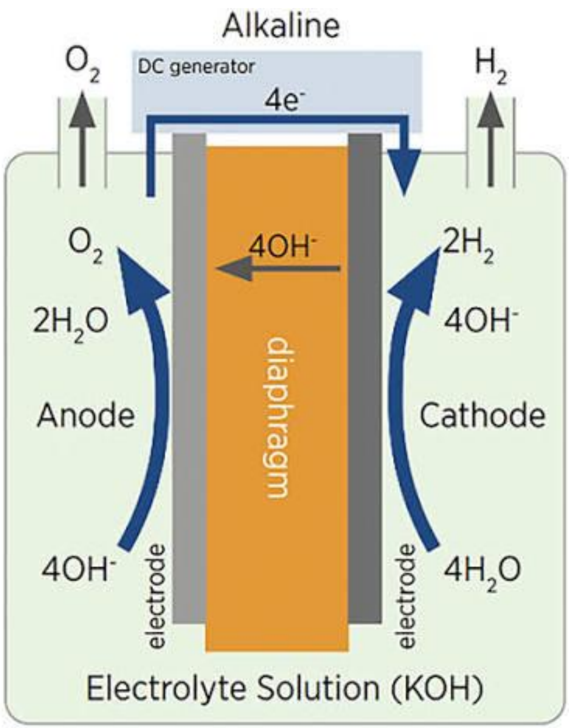
❑ Alkaline Water Electrolysis

Characterized by having two electrodes operating in *a liquid alkaline electrolyte*, where a 25–40 wt% KOH or NaOH solution is used

- A highly reliable, mature electrolysis technology with a technology readiness level (TRL) of 9
- High power consumption
- Necessity of corrosion-resistant electrodes (e.g., Ni-based alloy)
- Periodic replenishment of electrolyte



Science, **370**(6513) (2020) aba6118



Anode: $4OH^- \leftrightarrow 2H_2O+O_2+4e^-$

Cathode: $4H_2O+4e^- \leftrightarrow 2H_2+4OH^-$

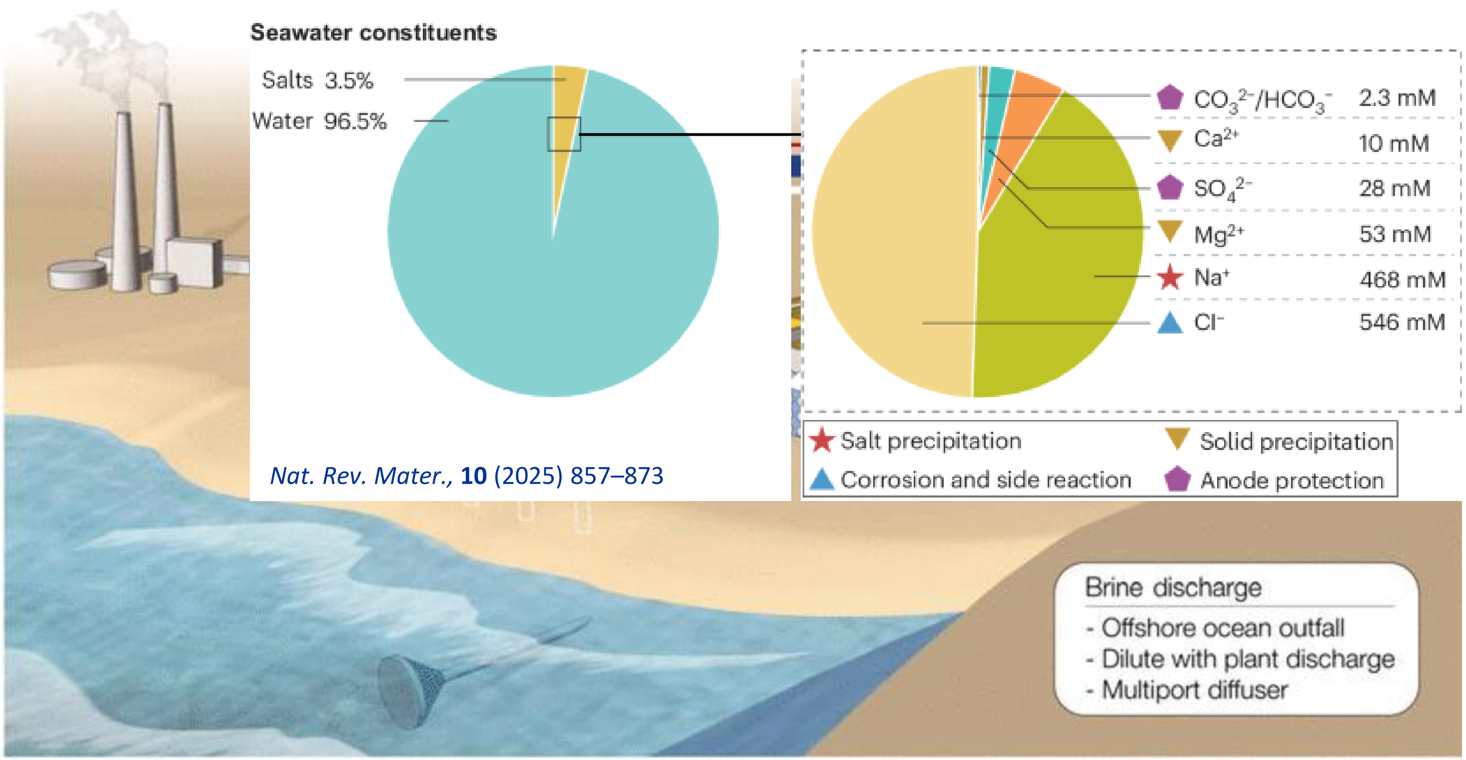
<Source: IRENA>

Electrolysis Hydrogen

❑ Alkaline Water Electrolysis

Significant interest in direct seawater electrolysis (saline water electrolysis) using ALKs

- Annually 4.43 Gt (=~12 Mt per day) of H_2O is required to produce 492 Mt of H_2 (2050 target)
- Abundant seawater (~97%) vs. extremely scarce high-purity freshwater (<1% of total 13.9 Gt)

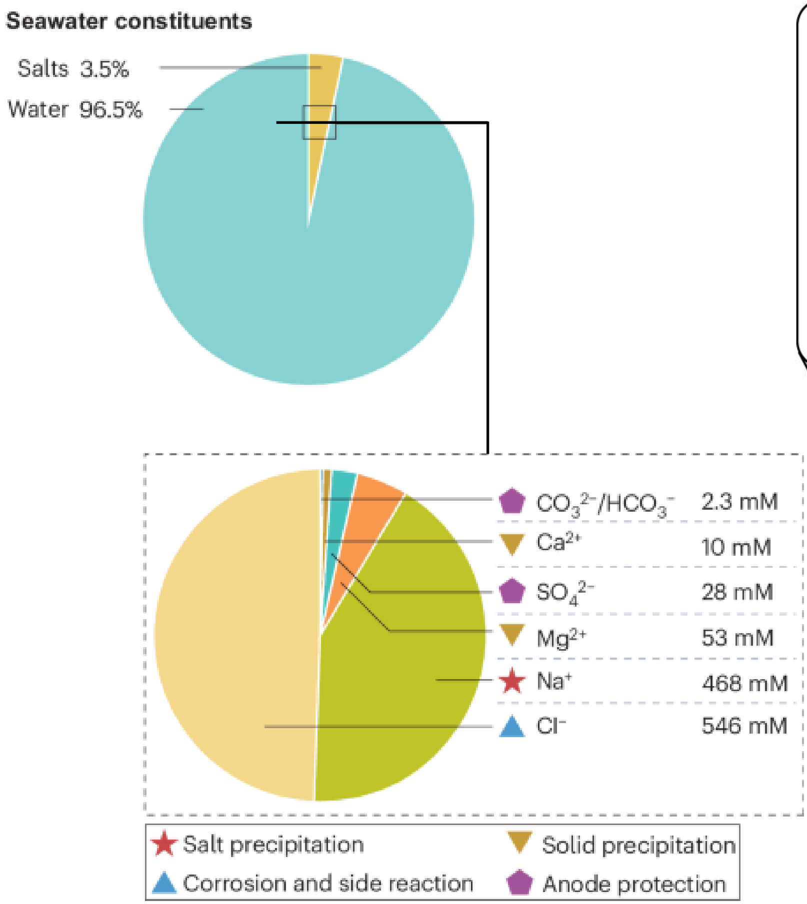


Science., **333**(6043) (2011) 712–717

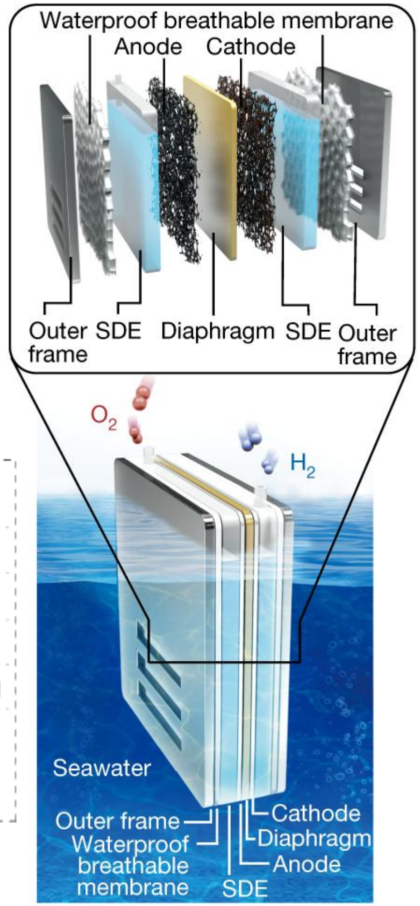
Electrolysis Hydrogen

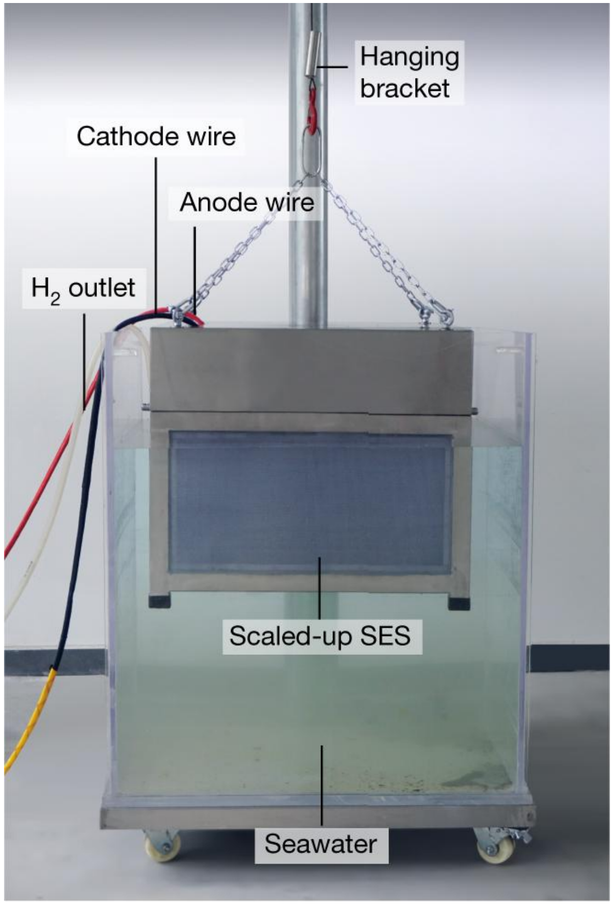
❑ Alkaline Water Electrolysis

Significant interest in direct seawater electrolysis (saline water electrolysis) using ALKs



Nat. Rev. Mater., **10** (2025) 857–873





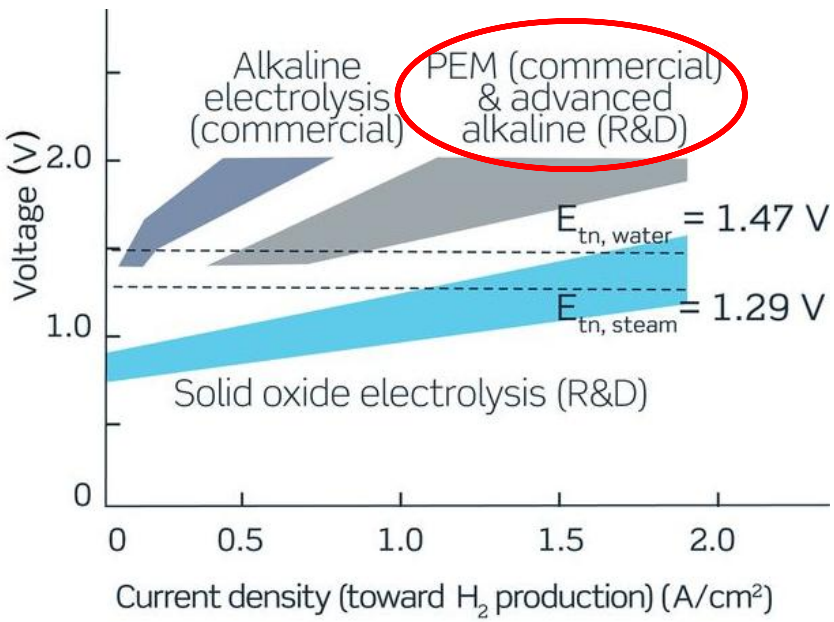
Nature, **612** (2022) 673–678

Electrolysis Hydrogen

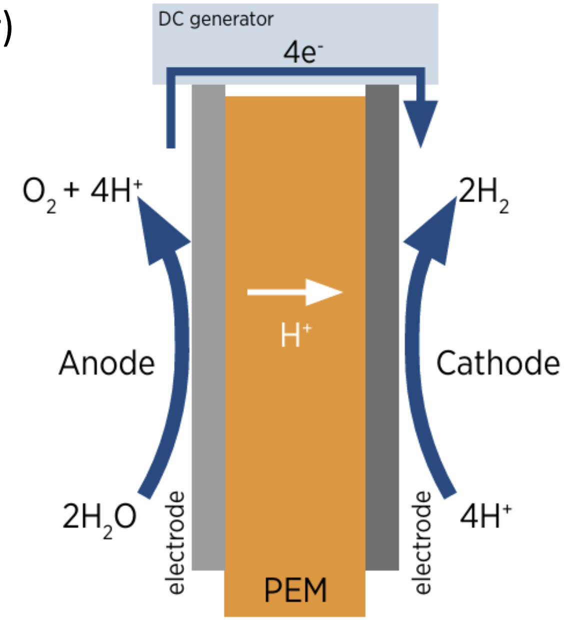
❑ Proton Exchange Membrane Water Electrolysis

Uses *a solid polymer electrolyte* to split pure H_2O into H_2 and O_2 using electricity

- A highly reliable, mature electrolysis technology with a technology readiness level (TRL) of 9
- Polymer membrane serving for ion exchange and stable electrolyte
- Simple, noncorrosive cell structure
- Highly dependent on a noble metal catalyst (e.g., Ir)



Science, **370**(6513) (2020) aba6118



Anode: $2H_2O \leftrightarrow O_2 + 4H^+ + 4e^-$

Cathode: $4H^+ + 4e^- \leftrightarrow 2H_2$

<Source: IRENA>

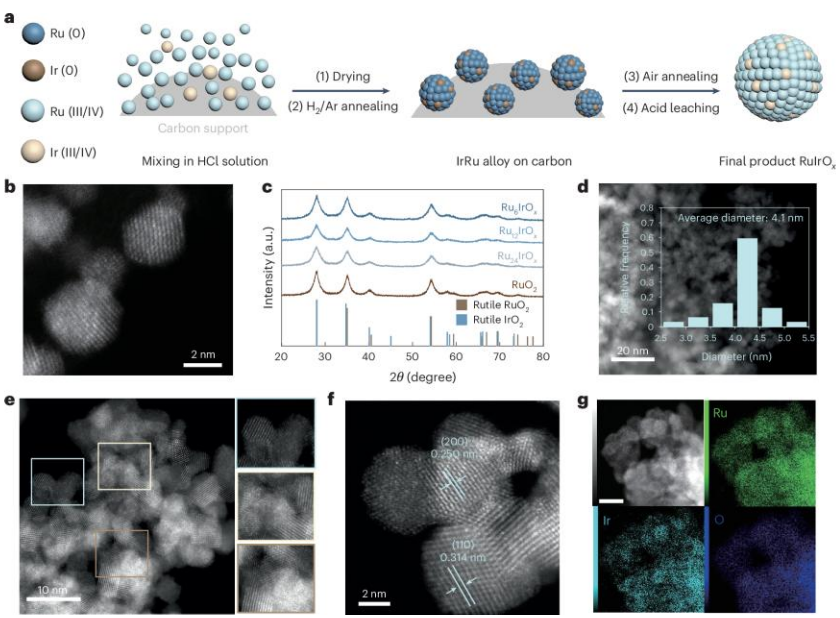
Electrolysis Hydrogen

❑ Proton Exchange Membrane Water Electrolysis

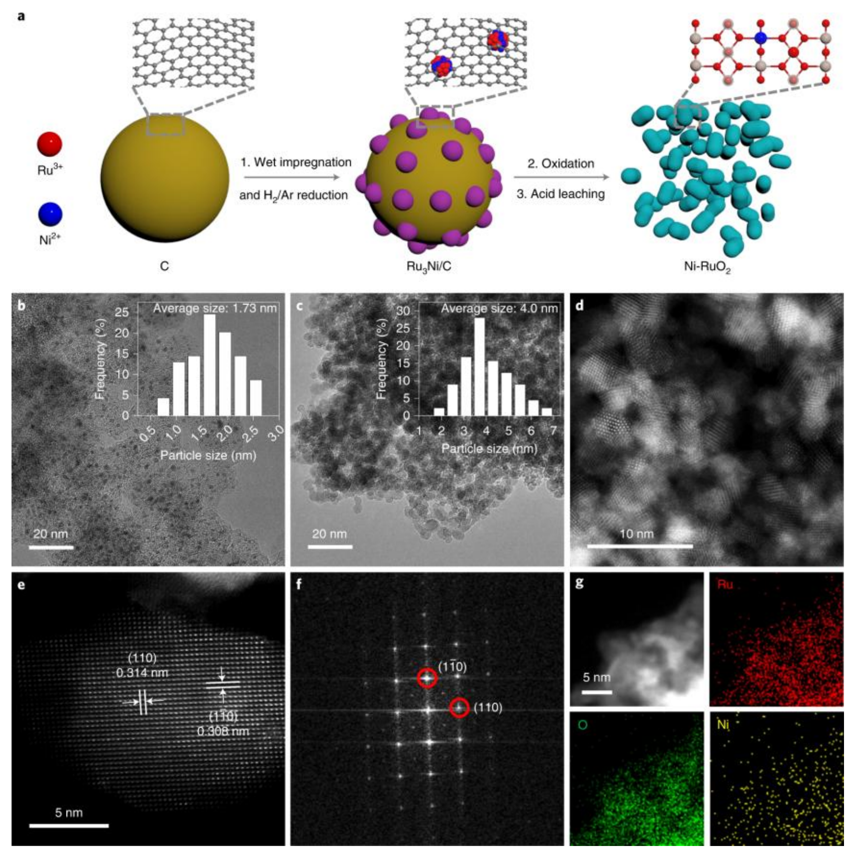
Significant interest in reducing dependence on expensive *iridium catalysts*

Table Prices of noble metals as of 2025 (\$ kg^{-1})

Ir	**144,000**	Ru	10,400
Au	75,430	Rh	147,000
Pt	27,800	Pd	49,500



Nat. Nanotechnol., **20** (2025) 1787–1795



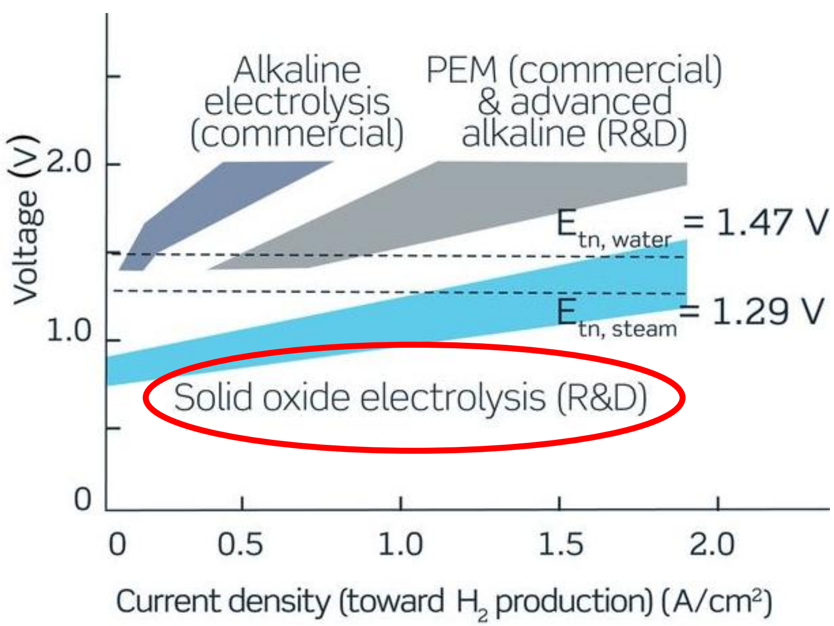
Nat. Mater., **22** (2023) 100–108

Electrolysis Hydrogen

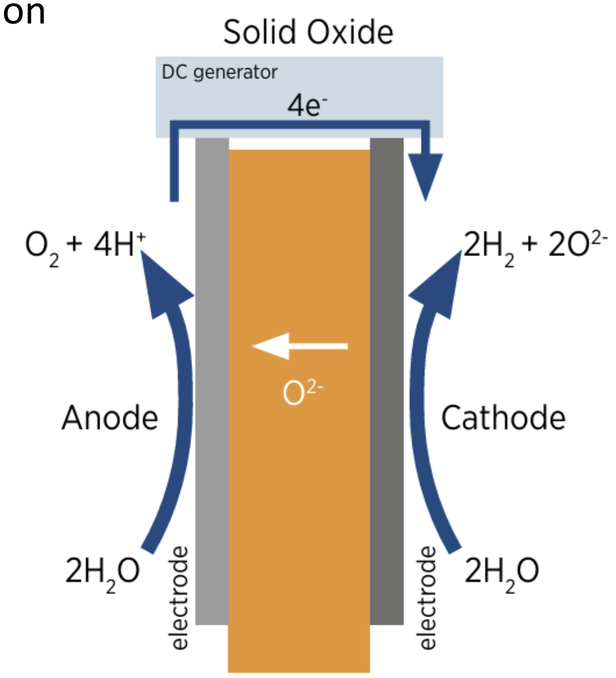
❑ Solid Oxide Electrolysis

Uses *a solid oxide, or ceramic, electrolyte* to produce H_2 gas (and/or CO) and O_2

- Highly efficient device due to high-temperature operation (>600 °C)
- Enables producing high-purity hydrogen
- Low competitiveness due to massive H_2 consumption



Science, **370**(6513) (2020) aba6118



Anode: $2O^{2-} \leftrightarrow O_2 + 4e^-$

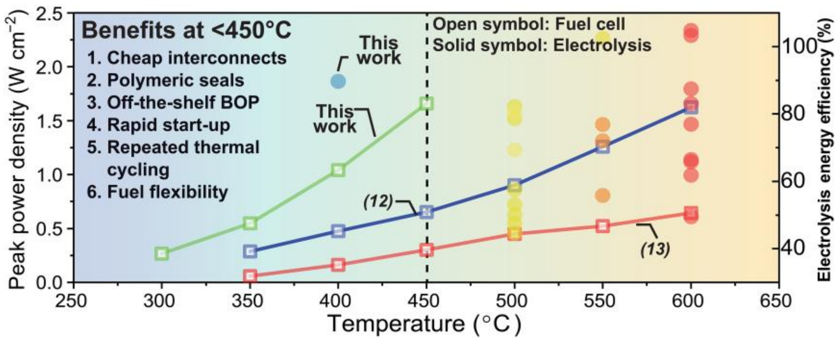
Cathode: $2H_2O + 4e^- \leftrightarrow 2H_2 + 2O^{2-}$

<Source: IRENA>

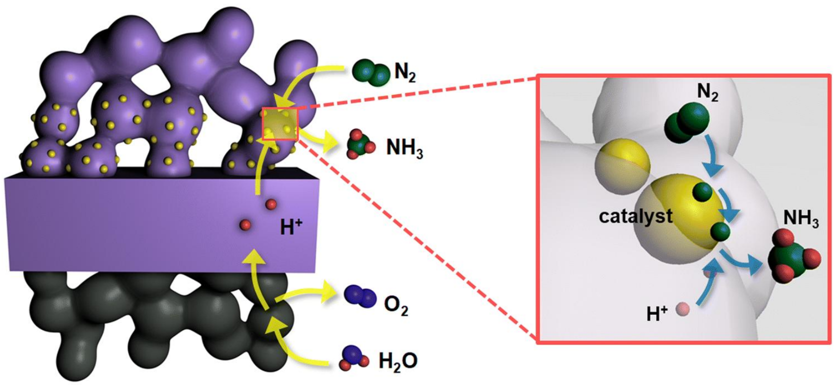
Electrolysis Hydrogen

❑ Solid Oxide Electrolysis

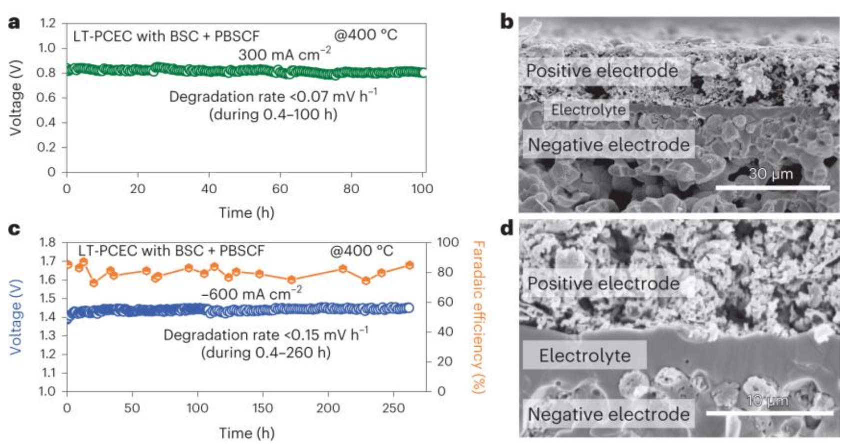
Significant focus on *lowering operating temp.* and *producing value-added chemicals* (CH_4, NH_3, etc.)



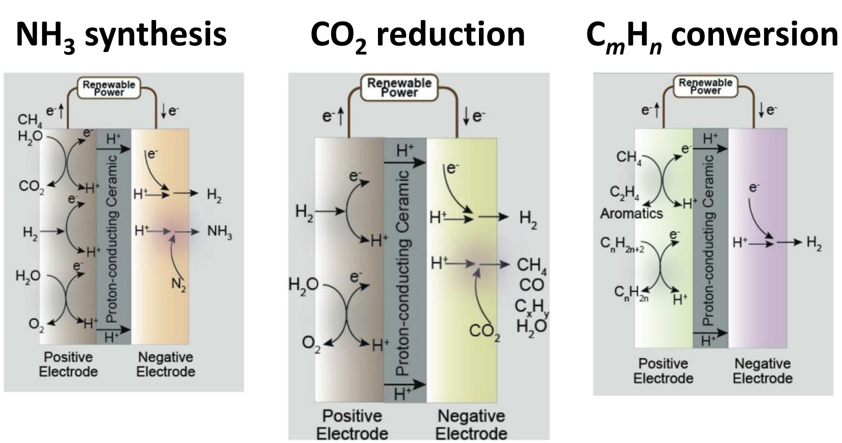
Sci. Adv., **11**(2) (2025) adq2507



J. Korean Ceram. Soc., **57** (2020) 480–494



Nat. Energy, **8** (2023) 1145–1157



Adv. Sci., **10** (2023) 2206478

Electrolysis Hydrogen

❑ Comparison of Electrolysis Technology

	ALK	PEM	SO
Operating temp. (°C)	70–90	50–80	**700–850**
Operating pressure (bar)	1–30	<70	**1**
Electrolyte	5–7 mol L^{-1} KOH	PFSA membrane	**Y_2O_3-stabilized ZrO_2 (YSZ)**
Separator	ZrO_2 stabilized with PPS mesh	Solid electrolyte (above)	**Solid electrolyte (above)**
Oxygen electrode	Ni-coated perforated stainless steel	IrO_2	**Perovskite oxide (e.g., LSCF, LSM)**
Hydrogen electrode	(above)	Pt/C	**Ni–YSZ**
Porous transport layer (Anode)	Ni mesh	Pt-coated sintered porous Ti	**Ni mesh**
Porous transport layer (Cathode)	Ni mesh	Sintered porous Ti or C cloth	**–**
Bipolar plate (Anode)	Ni-coated stainless steel	Pt-coated Ti	**–**
Bipolar plate (Cathode)	Ni-coated stainless steel	Au-coated Ti	**Co-coated stainless steel**
Frames and Sealing	PSU, PTFE, EPDM	PTFE, PSU, ETFE	**Ceramic glass**

<Source: IRENA>

Electrolysis Hydrogen

❑ Comparison of Electrolysis Technology (Today vs. 2050)

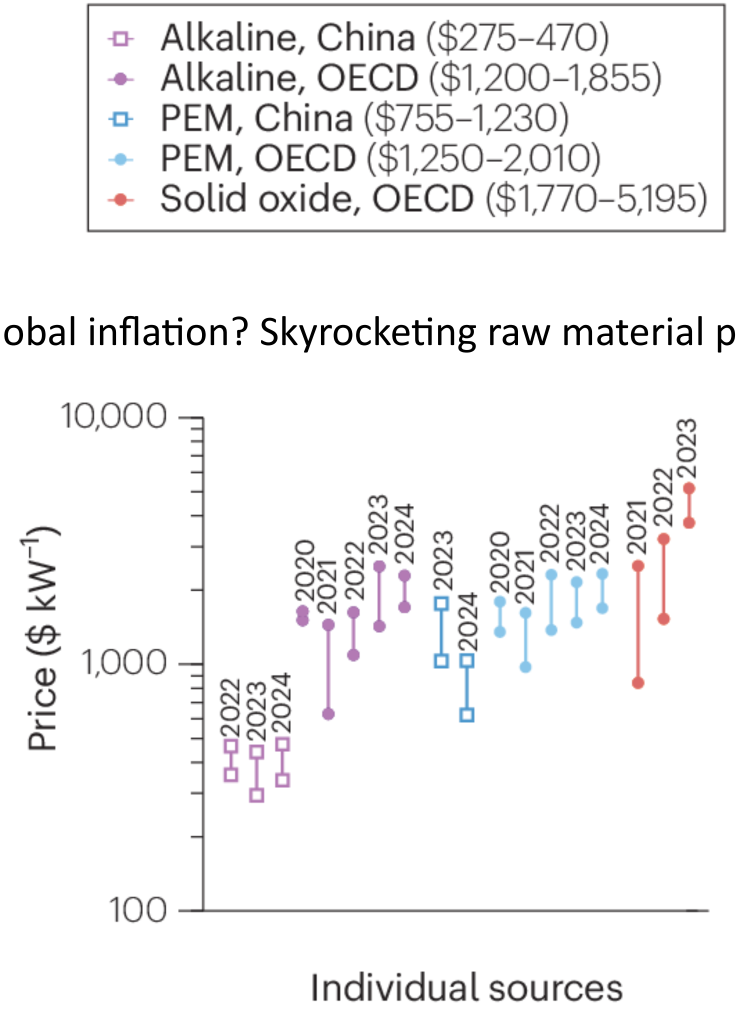
	2020			2050		
	ALK	PEM	**SOEC**	ALK	PEM	**SOEC**
Nominal current density (A cm^{-2})	0.2–0.8	1–2	**0.3–1**	>2	4–6	**>2**
Voltage range (V)	1.4–3.0	1.4–2.5	**1.0–1.5**	<1.7	<1.7	**<1.48**
Operating temp. (°C)	70–90	50–80	**700–850**	>90	80	**<600**
Cell pressure (bar)	<30	<30	**1**	>70	>70	**>20**
H_2 purity (%)	>99.9	>99.9	**99.9**	>99.9999	Same	**>99.9999**
Voltage eff. (LHV) (%)	50–68	50–68	**75–85**	70	>80	**85**
Electrical eff. (stack) (kWh kg^{-1})	47–66	47–66	**35–50**	<42	<42	**<35**
Electrical eff. (system) (kWh kg^{-1})	50–78	50–83	**40–50**	<45	<45	**<40**
Lifetime (system) (kh)	60	50–80	**<20**	100	100–120	**80**
Stack unit size (MW)	1	1	**0.005**	10	10	**0.2**
Electrode area (cm^2)	>10000	1500	**200**	30000	>10000	**500**
Startup (to nominal load) (min)	<50	<20	**>600**	<30	< 5	**<300**
CAPEX (stack) (>1 MW) ($ kW^{-1})	270	400	**>2000**	<100	<100	**<200**
CAPEX (system) (>10 MW) ($ kW^{-1})	500–1000	700–1400	**N/A**	<200	<200	**<300**

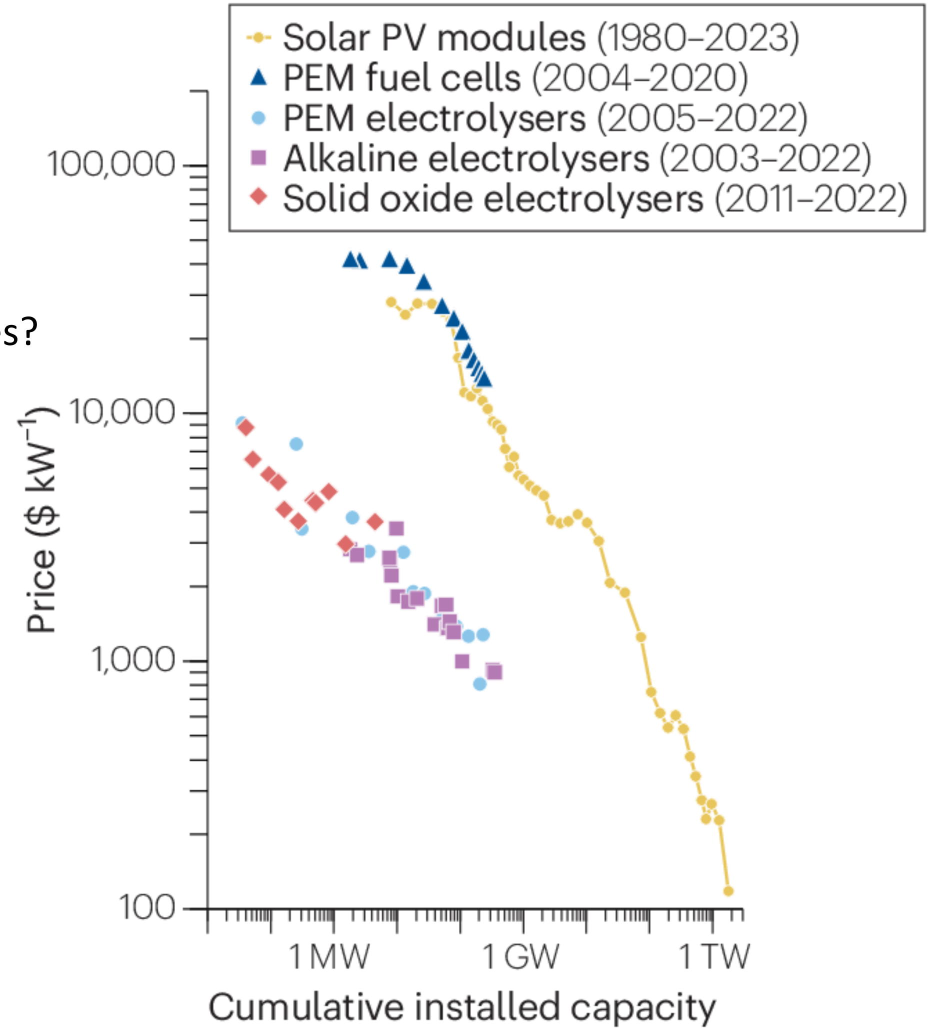
<Source: IRENA>

Electrolysis Hydrogen

❑ Levelized Cost of Hydrogen (LCOH) & Capital Cost (CAPEX)

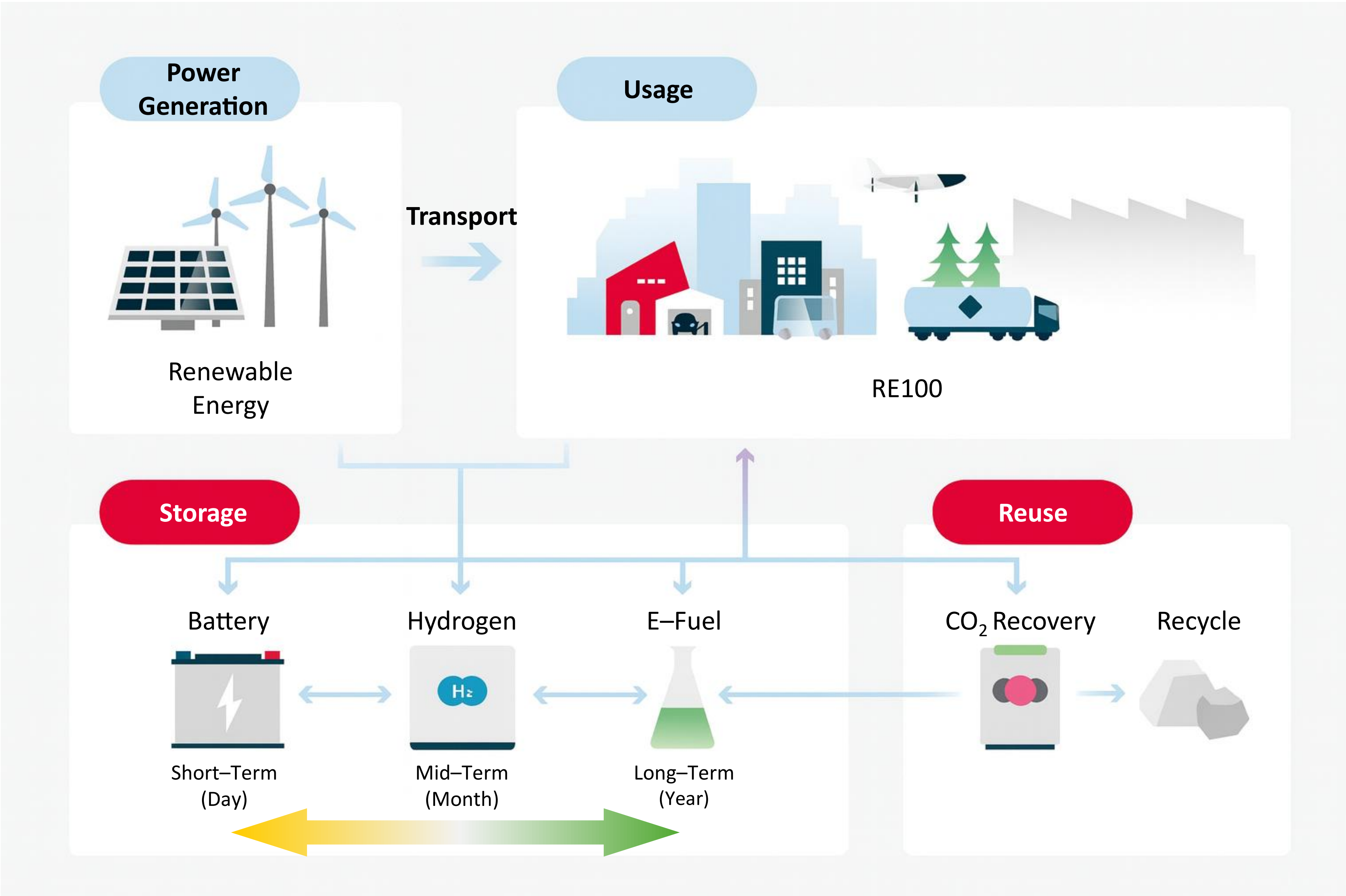
Global inflation? Skyrocketing raw material prices?





Nat. Rev. Clean Technol., **1** (2025) 351–371

Surplus RE Hydrogen Utilization



Key Technological Objectives

❑ Produce, store/transport, and use

Establishing, standardizing, and commercializing secure, low–cost technologies across these *three primary pillars* is essential for introducing a full hydrogen society

PRODUCE

- Low cost
- Large scale
- Carbon–free production

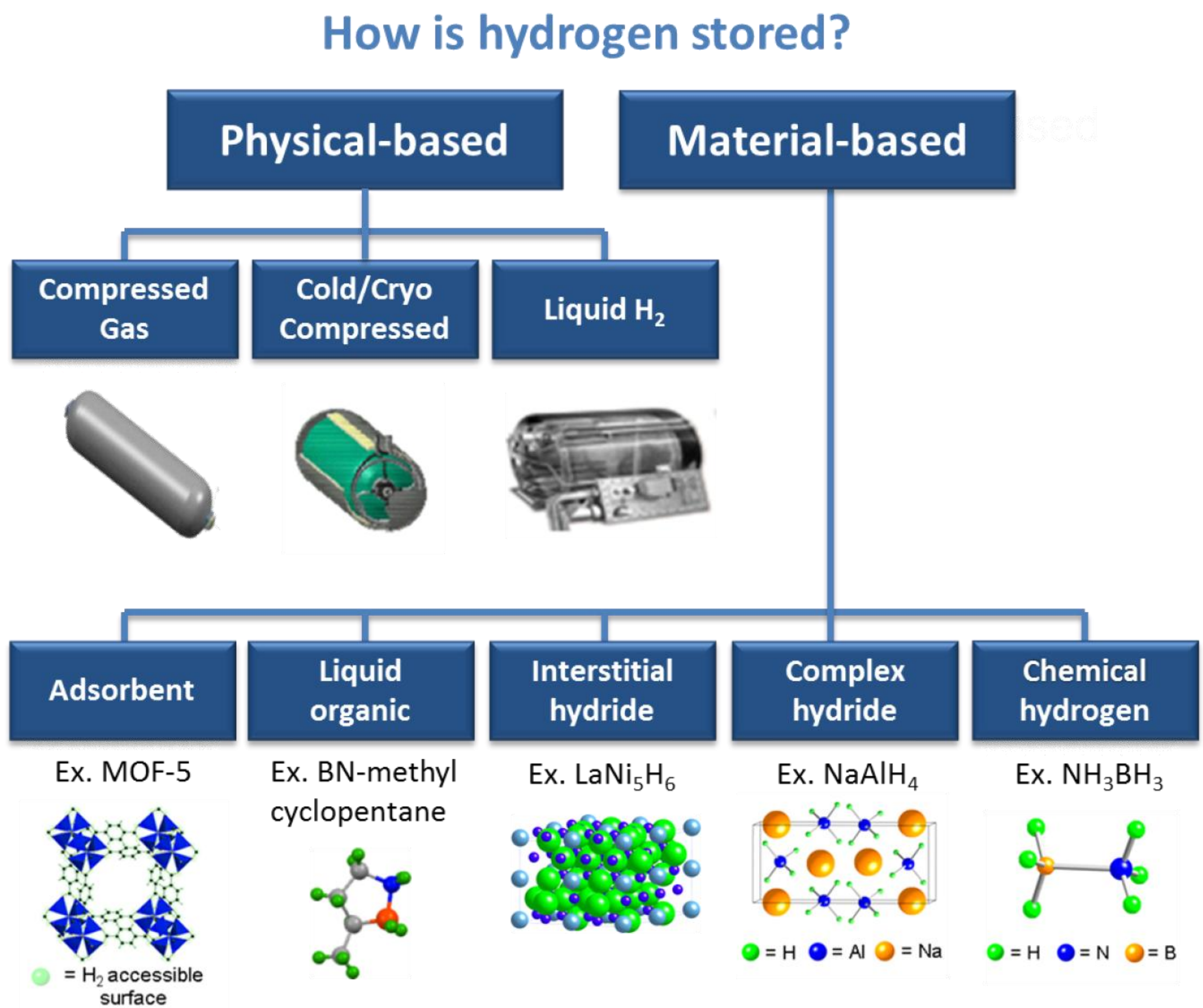
STORE & TRANSPORT

- High density
- Loss free
- Safe contaminant

USE

- High efficiency
- Clean energy
- Safe energy

Hydrogen Storage Technology



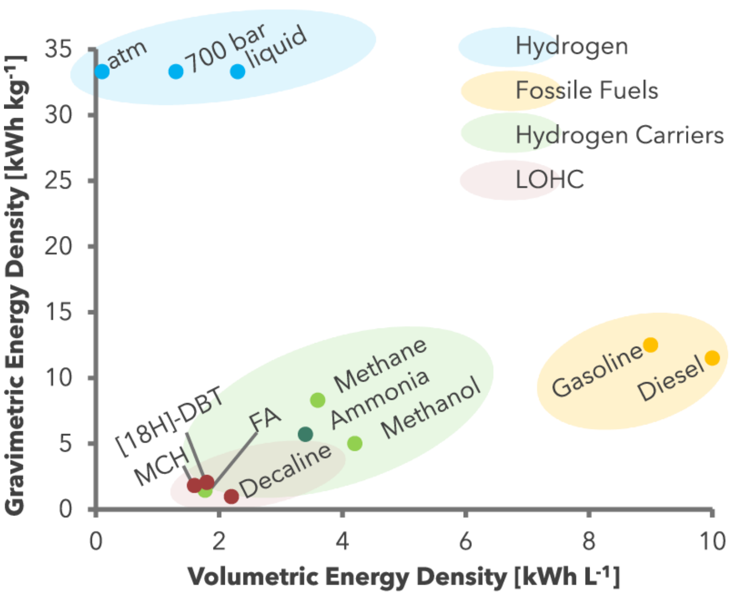
<Source: US Department of Energy>

Hydrogen Storage Technology

❑ Energy Density of H_2

H_2 has high mass energy density, but *low volumetric energy density* at STP (0 °C, 1 atm)

- **Mass energy density**
  241.8 kJ mol^{-1} ÷ 0.002016 kg mol^{-1} = 119.9 MJ kg^{-1} (cf. Gasoline: ~44 MJ L^{-1})
- **Volumetric energy density**
  241.8 kJ mol^{-1} ÷ 22.4 L mol^{-1} = 0.01079 MJ L^{-1} (cf. Gasoline: ~32 MJ L^{-1})



Nat. Comm., **15** (2024) 7268

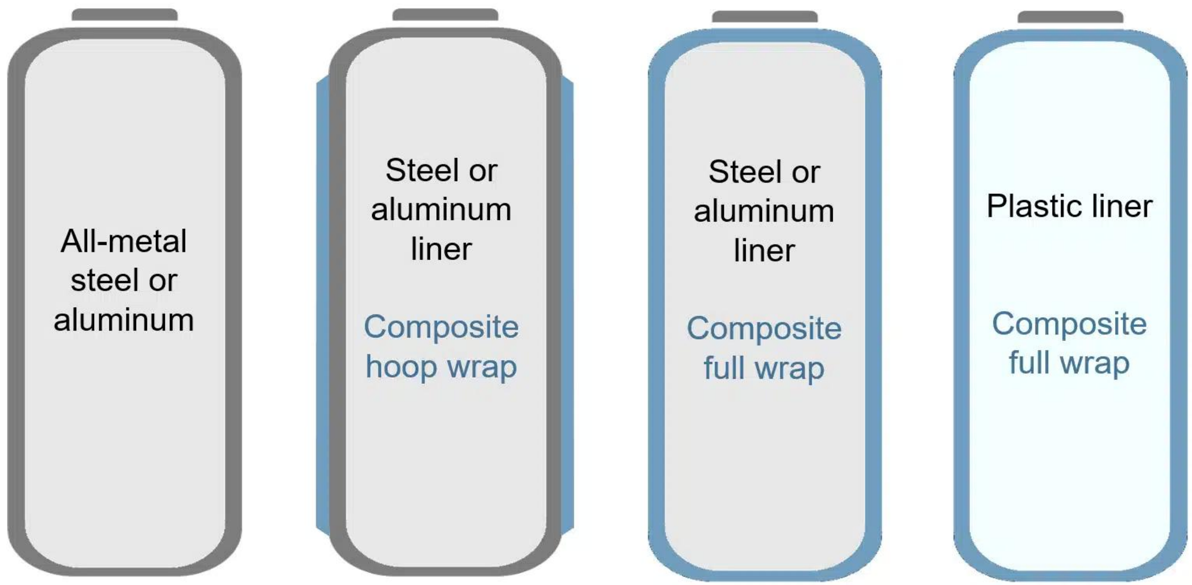
Table Volume and density of compressed H_2

Pressure (bar)	Normalized volume (%)	Density (kg m^{-3})	Remarks
1.0	100	0.0887	Ambient
100	1.1	8.32	Industry
200	0.57	15.6	Tube trailer
350	**0.35**	**24.9**	**Heavy-duty**
700	**0.21**	**41.1**	**FCV**
900	0.18	47.9	H_2 station

Physical-Based H_2 Storage

❑ Compressed Gaseous H_2 (CGH_2)

Compressed hydrogen in tanks at 350 and 700 bar is used for mobile hydrogen storage in FCVs



	Type I	Type II	Type III	Type IV
Pressure	200 bar	<300 bar	350–700 bar	350–700 bar
Relative Weight	3.3	2.7	1.4	1.0
Relative Capacity	30%	35%	75%	100%
Cost	Lowest	Low	High	Slightly high *\$15 kWh^{-1} (=\$500 kg_{H2}^{-1}) (2015)
Stability	Low	Low	High	High

Physical-Based H_2 Storage

❑ Hydrogen Truck (Tube Trailor)

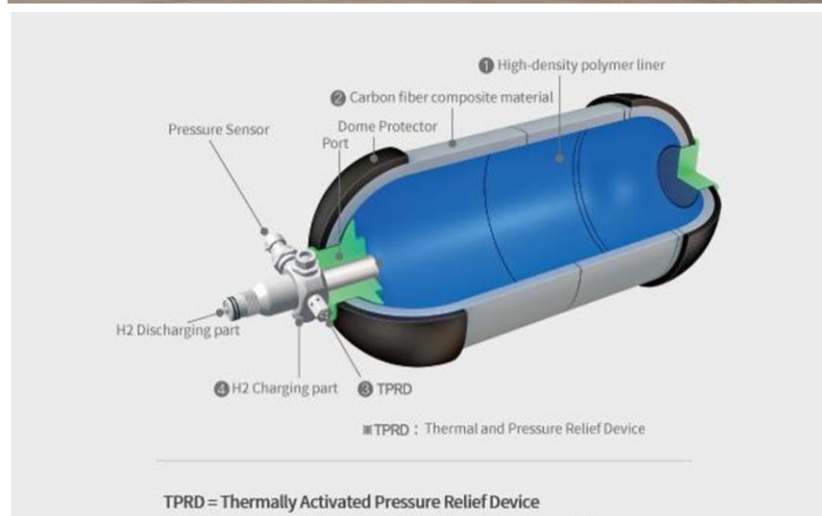
Trucks that haul gaseous hydrogen, which is typically compressed to 180 bar or higher into long cylinders stacked on the trucks.

- **Energy for compression**
  Typically, ~2–4.4 kWh kg_{H2}^{-1}
- **Cost for compression**
  Depends on distance, electricity price, and pressure

Hydrogen		타입1(Type1) 수소 튜브트레일러	타입4(Type4) 수소 튜브트레일러
수소 튜브트레일러란?		· 특장차에 기체 수소 저장 용기가 장착된 차량을 의미 · 기체 수소 저장 용기에 따라 구분	
기체 수소 저장 용기	재질	· 타입1 저장용기 사용 · 용기 전체가 금속재질 라이너(내측)로 구성	· 타입4 저장용기 사용 · 플라스틱과 같은 비금속 라이너(내측)에 탄소섬유 복합재료로 용기 전체를 보강한 형태
	압력	· 주로 200bar 사용하나, · 150~300bar까지 저장 가능	· 제조사별 상이하나, · 350~700bar까지 저장 가능
수송 용량		· 최대 수송용량: 340kg · 수송 튜브트레일러 중량 대비 수소 저장량이 작음	· 최대 수송용량: 634kg~1,000kg(개발 중) · 기존 대비 운송비 절감

<Source: KPMG>


<Source: Department of Energy (DOE)>



Eur. J. Energy Res., **5**(2) (2025)

Physical-Based H_2 Storage

❑ Hydrogen Pipeline

Used to connect the point of hydrogen production or delivery of hydrogen with the point of demand

- **As of 2024**

  USA – 2,600 km of low pressure hydrogen pipelines

  Europe – 1,600 km of low pressure hydrogen pipelines

  <Source: EPRI, World Economic Forum>

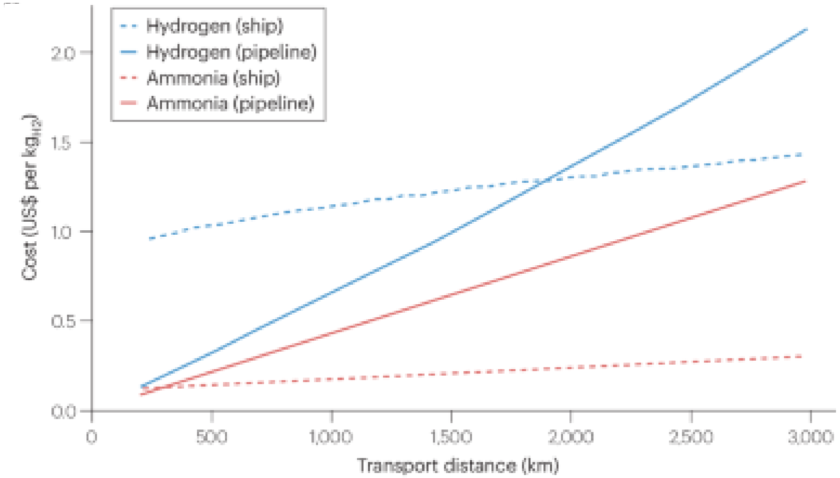
- **Price**

  $~0.5–1.0 kg_{H2}^{-1} per 1,000 km

- **Weakness**

  Requires massive initial investment and dedicated special piping due to *high leakage risks*, *metal embrittlement*, and *a wide flammability range*.

<Source: ABC News>



Nat. Comm., **15** (2024) 7268

Physical-Based H_2 Storage

❑ Liquified H_2 (LH_2)

The liquid state of hydrogen, which is commonly used as a liquid rocket fuel for rocketry application

*NASA, US Air Force, etc.

- **Low boiling point**

  –252.9 °C (=20.3 K) at atmospheric pressure (1 atm)

  Liquefaction consumes more than 30% of the energy content of the hydrogen and is expensive.

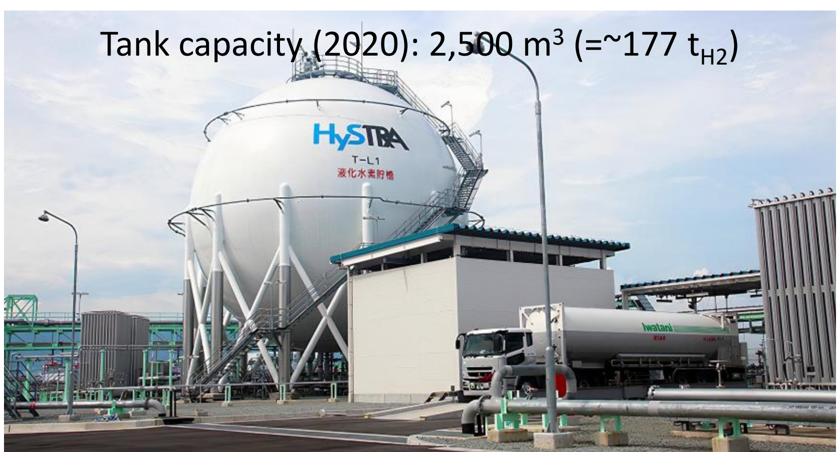
  **Theoretically, H_2 requires ~3.4 kWh kg^{-1} to liquify

  ***Existing liquefaction facilities use 10–13 kWh kg^{-1} (LHV 33 kWh kg^{-1})

- **Higher density**

  Its density is 70.9 g L^{-1}, which is ~790 and ~1.73 times higher than those of atmospheric H_2 (0.090 g L^{-1}) and compressed H_2 at 700 bar (41.1 g L^{-1}).

  But it is still light (relative density = 0.07), compared to gasoline (0.74) and LNG (0.45).



Tank capacity (2020): 2,500 m^3 (=~177 t_{H2})

<Source: Iwatani Corp.>

Tank capacity (2030): 80,000 m^3 (=~5,700 t_{H2})

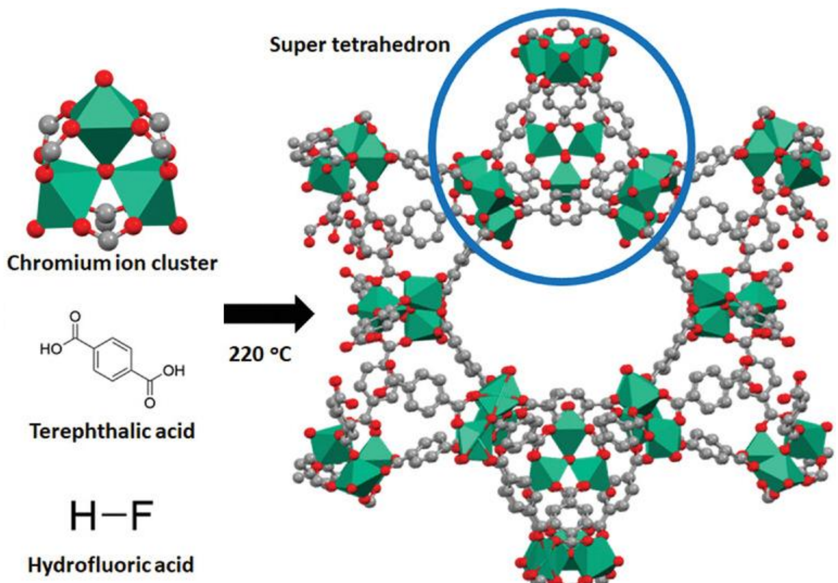
<Source: HD Hyundai>

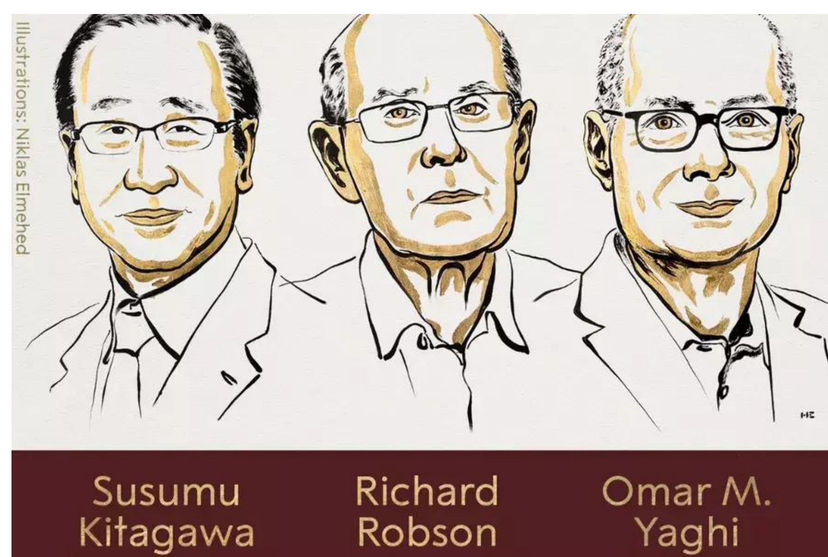
Material-Based H_2 Storage

❑ Metal–Oxide Framework (MOF)

A method of storing hydrogen molecules via physical adsorption within ultra-porous 3D crystalline structures composed of metal ions and organic linkers

- **Fast Kinetics & High Capacity**
  High gravimetric capacity (7–14 wt%) at moderate pressures (30–100 bar) with fully reversible, rapid charge/discharge
  *MOF + xH_2 ⇔ MOF • xH_2 + Heat (ΔH = ~–4–10 kJ mol^{-1})
- **Cryogenic Dependency**
  Low storage capacity at RT due to poor adsorption efficiency, requiring cryogenic cooling (~77 K)





Material-Based H_2 Storage

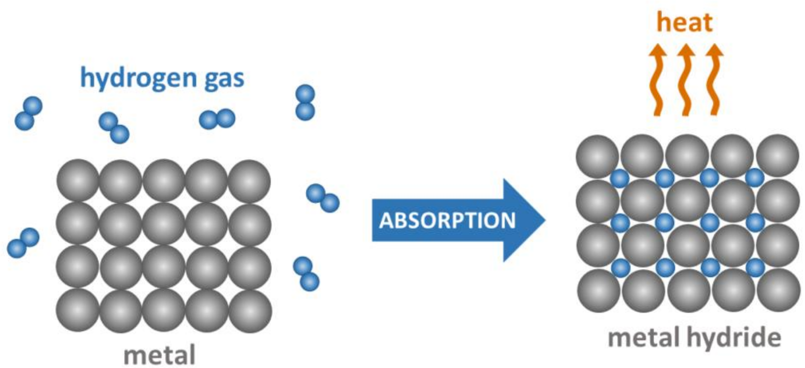
❑ Metal Hydride

A method of storing in the form of compounds with light metals (such as Li, B, Mg, and Al) or carbon (such as activated carbon and carbon nanotubes)

- **Safety**
  Operates at low pressures (often <10–50 bar) with high volumetric capacity (~100 kg m^{-3}) and low explosion risks compared to high-pressure gas tanks

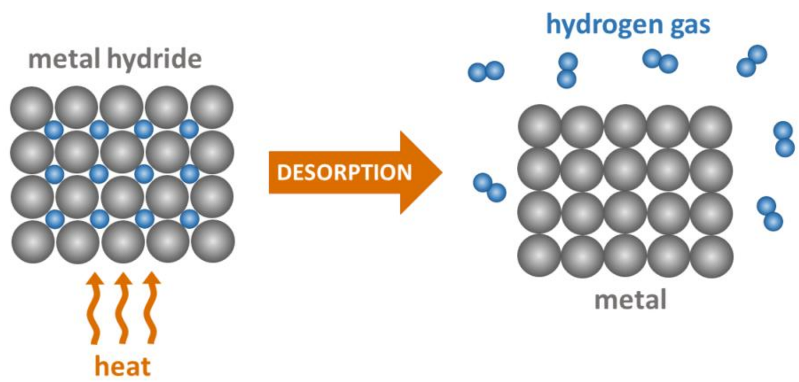
- **Heavy Weight**
  Low gravimetric capacity (~1–2 wt%), making it heavy for mobile transportation

Exothermic Reaction



$2Me + xH_2 \rightarrow 2MeH_x + Heat$
(Me = La, Ti, Mg, Al, etc.)

Endothermic Reaction



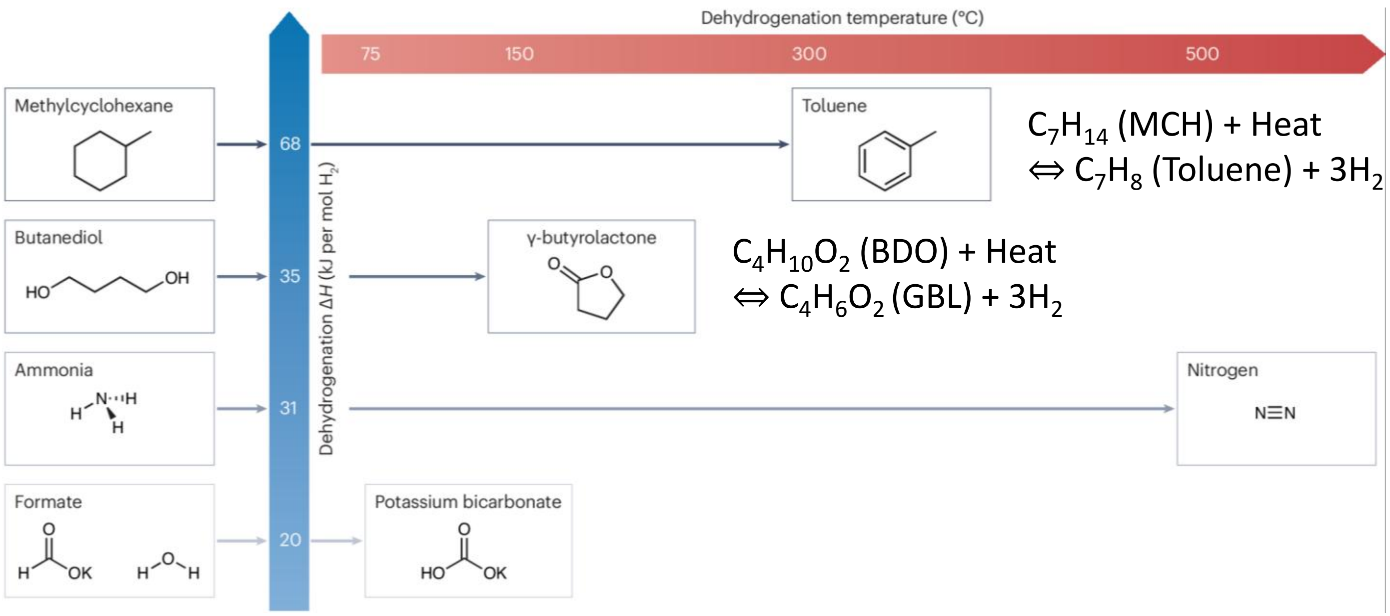
$2MeH_x + Heat \rightarrow 2Me + xH_2$
(Me = La, Ti, Mg, Al, etc.)

Material-Based H_2 Storage

❑ Liquid Organic Hydrogen Carrier (LOHC)

A technology that allows hydrogen to be chemically bonded to a liquid carrier (e.g., organic oil), making it safer and more efficient to handle

- **Uses Petroleum Infrastructure**
  Liquid at ambient condition; 100% compatible with existing tankers and oil pipelines
- **High Heat Demand**
  Requires high-temperature heat input (~250–300 °C) and noble catalysts for hydrogen release



C_7H_{14} (MCH) + Heat
$\Leftrightarrow$ C_7H_8 (Toluene) + $3H_2$

$C_4H_{10}O_2$ (BDO) + Heat
$\Leftrightarrow$ $C_4H_6O_2$ (GBL) + $3H_2$

Nat. Rev. Clean Technol., **2** (2026) 512–533

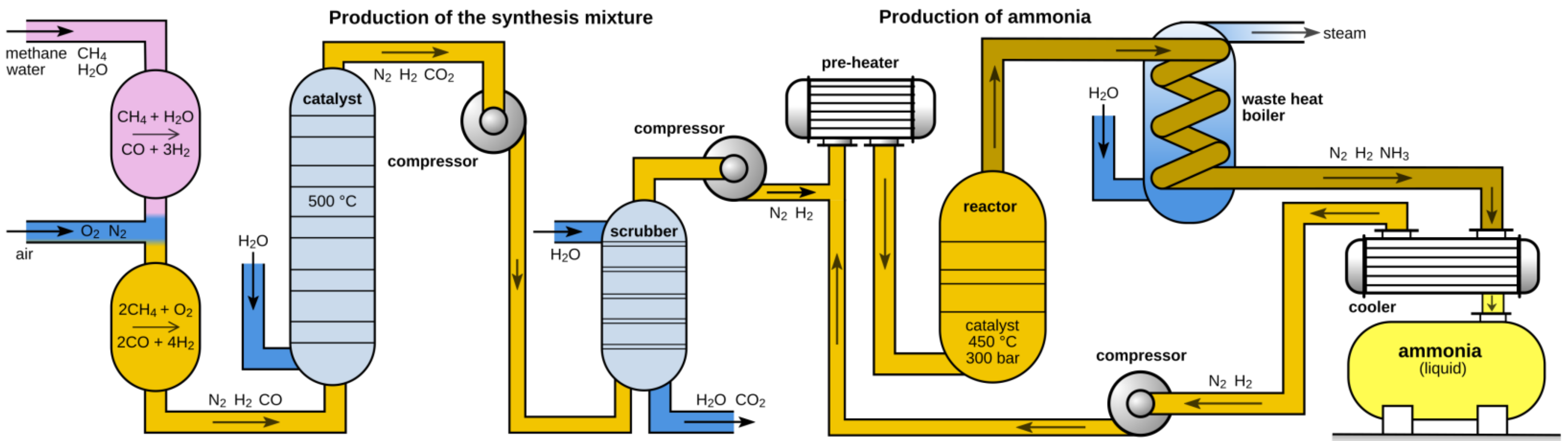
Material-Based H_2 Storage

❑ Ammonia (NH_3)

A technology that allows hydrogen to be chemically bonded to a liquid carrier (e.g., organic oil), making it safer and more efficient to handle

- **Highest Hydrogen Density**
  Delivers high gravimetric capacity (17.7 wt%) with zero carbon emissions while utilizing mature global ammonia shipping infrastructure
- **Toxicity & High Cracking Energy Demand**
  Requires strict safety protocols due to toxicity and high thermal energy input (500 °C) for endothermic cracking into pure H_2

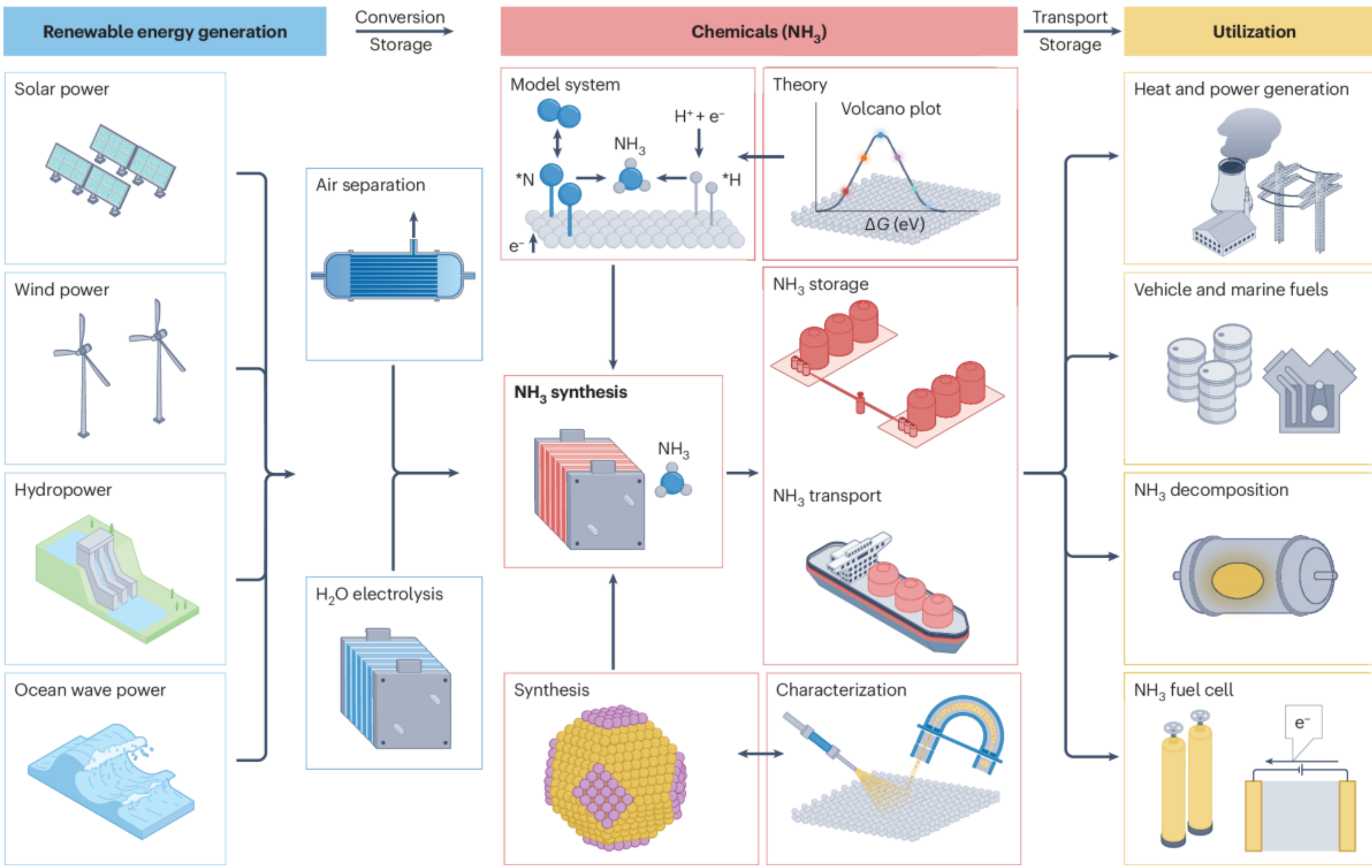
$N_2 + 3H_2 \Leftrightarrow 2NH_3$ ($\Delta H° = 92.3$ kJ mol^{-1})



<Source: https://en.wikipedia.org/wiki/Haber_process>

Material-Based H_2 Storage

❑ Ammonia (NH_3)



Nat. Rev. Clean Technol., **1** (2025) 755–770

Hydrogen Storage Technology

❑ Technology Comparison

Technology	State	Volumetric density	Gravimetric density	Key advantage	Challenges
CGH_2	Gas (350–700 bar, Ambient)	Low (~40 kg m^{-3})	4.5–5.7 wt%	Fast refueling Mature tech	High pressure Vessel cost
LH_2	Liquid (1–10 bar, –253 °C)	High (~70 kg m^{-3})	11–20 wt%	Low volume	High energy Boil-off gas
MOF	Solid (30–100 bar, –77 °C)	High (~50–60 kg m^{-3})	7.0–10 wt%	Fast kinetics Low pressure	Cryogenic temp. Low RT capacity
Metal Hydride	Solid (10–50 bar, Ambient)	Very high (~100 kg m^{-3})	1.0–1.5 wt%	Low pressure Safety	Heavy mass Heat mgmt.
LOHC	Liquid (Ambient Pressure/Temp)	Medium (~60 kg m^{-3})	6.0–6.2 wt%	Uses existing infrastructure	High release heat
Ammonia	Liquid (8.5 bar or –33 °C)	Very high (~120 kg m^{-3})	17.7 wt%	High capacity Carbon-free	Toxicity Cracking cost

Key Technological Objectives

❑ Produce, store/transport, and use

Establishing, standardizing, and commercializing secure, low–cost technologies across these *three primary pillars* is essential for introducing a full hydrogen society

PRODUCE

- Low cost
- Large scale
- Carbon–free production

STORE & TRANSPORT

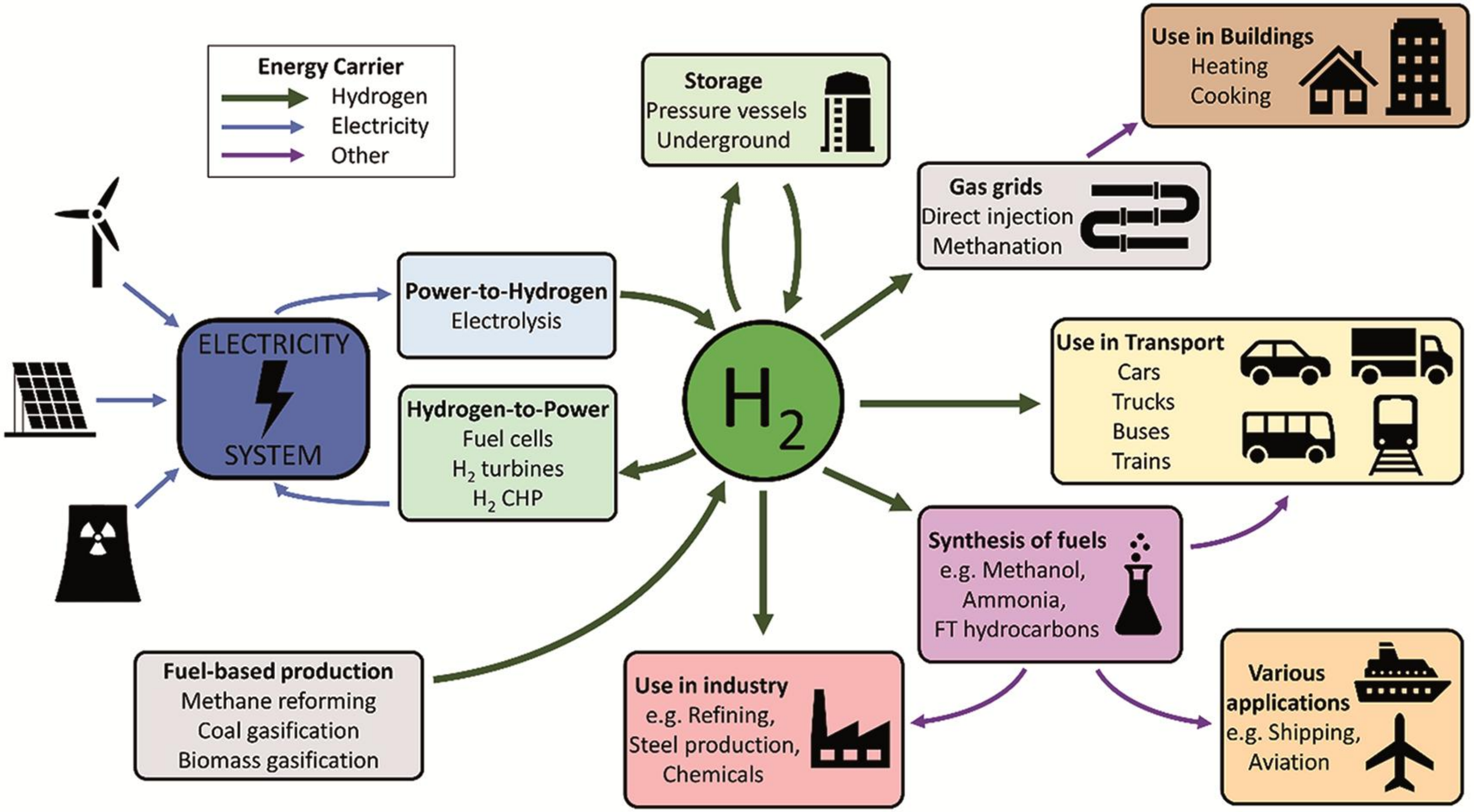
- High density
- Loss free
- Safe contaminant

USE

- High efficiency
- Clean energy
- Safe energy

Hydrogen Use

❑ Global Hydrogen Use by Sector

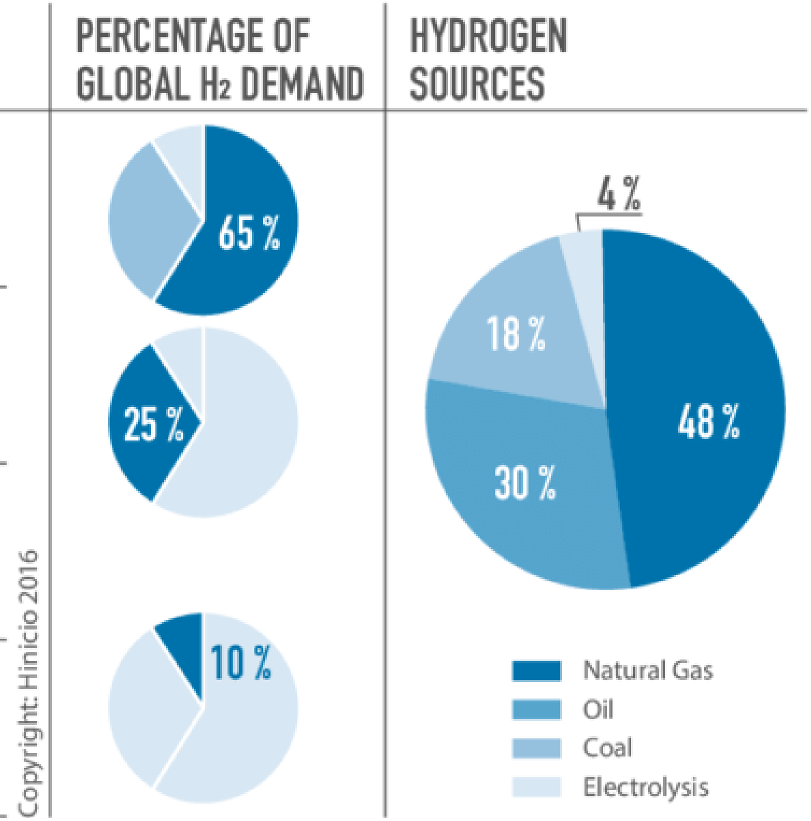


Sustain. Energy Fuels, **4** (2019) 80–95

Hydrogen Use

❑ Global Hydrogen Use by Sector

INDUSTRY SECTOR	KEY APPLICATIONS
CHEMICAL ~50%	• Ammonia • Polymers • Resins
REFINING ~43%	• Hydrocracking • Hydrotreating
IRON & STEEL ~5%	• Annealing • Blanketing gas • Forming gas
GENERAL INDUSTRY	• Semiconductor • Propellant fuel • Glass production • Hydrogenation of fats • Cooling of generators



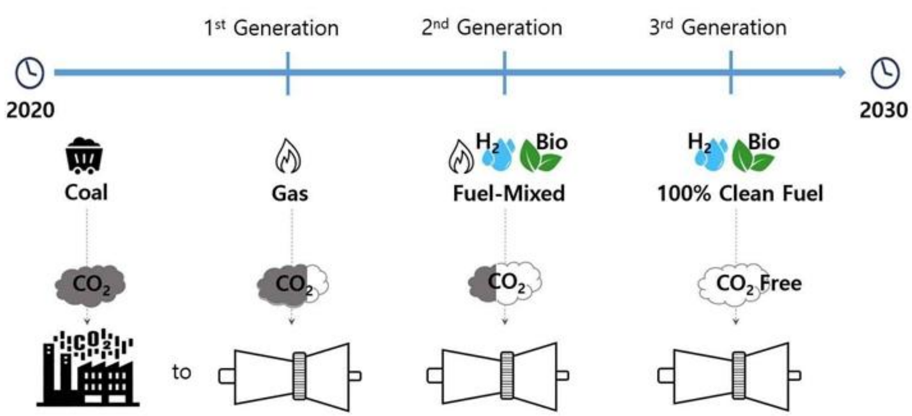
<Source: IRENA (2018)>

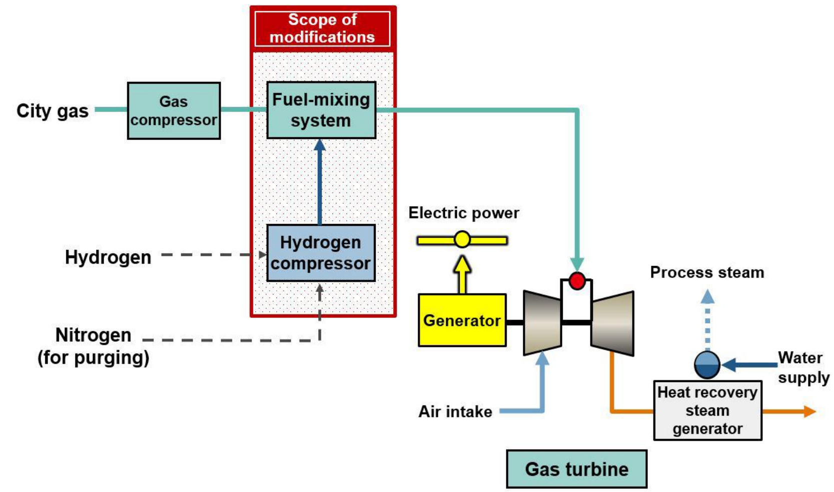
– Of the 100 Mt of H_2 produced in 2021, 43% was used in oil refining and 57% in industry, principally in the manufacture of NH_3 for fertilizers, and CH_3OH

Hydrogen Use

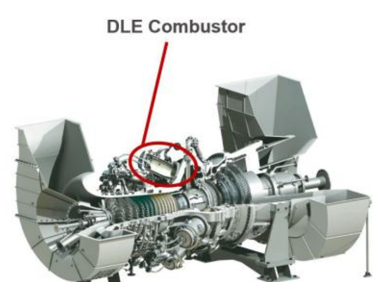
❑ Hydrogen Gas Turbine (GT)

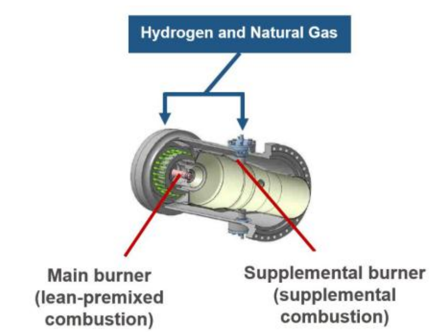
Unlike conventional GTs operated by LNG, hydrogen GTs emit zero CO_2 during power generation, as their combustion process releases only H_2O





Properties		Unit	Methane	Hydrogen
Formalr		-	CH_4	H_2
Molecular weight		g/mol	16	2
Density		kg/nm^3	0.716	0.090
Heating value (LHV)	Weight	MJ/kg	50	120
	Volume	MJ/nm^3	35.8	10.8





<Source: https://global.kawasaki.com/en/corp/newsroom/news/detail/?f=20220802_0328>

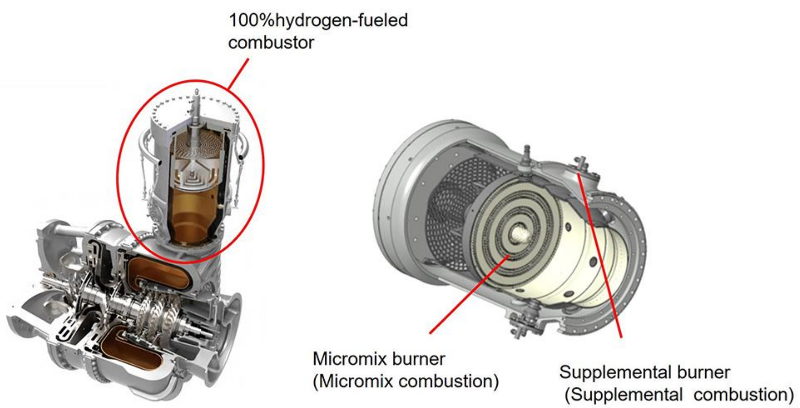
Hydrogen Use

❑ Hydrogen Gas Turbine (GT)

Unlike conventional GTs operated by LNG, hydrogen GTs emit zero CO_2 during power generation, as their combustion process releases only H_2O

Kawasaki Launches World's First 1.8 MW Class, 100% Hydrogen-fueled, Dry-combustion Gas Turbine Cogeneration System

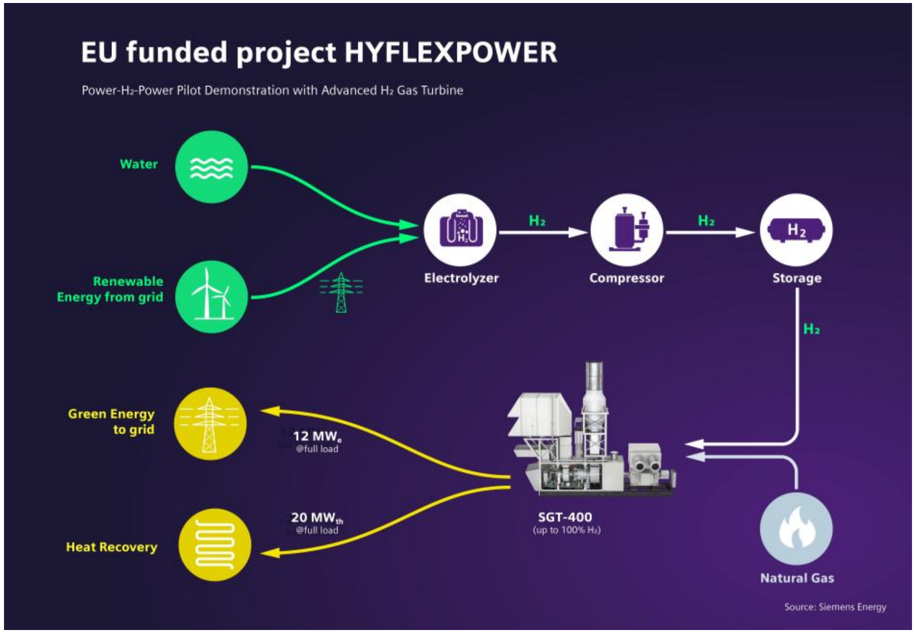
Sep. 05, 2023



<Source: https://global.kawasaki.com/en/corp/newsroom/news/detail/?f=20230905_2781>

Siemens Energy team demonstrates gas turbine running on 100% hydrogen

Oct 17, 2023, 11:12:02 AM | Article by Plamena Tisheva



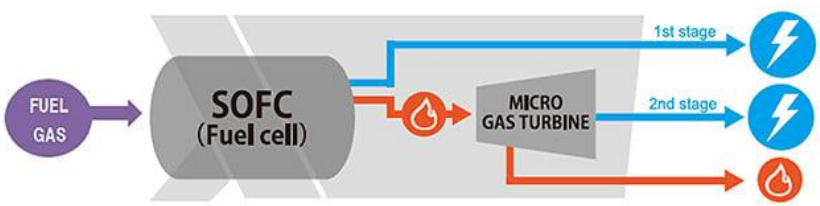
<Source: https://www.siemens-energy.com/global/en/home/press-releases/hyflexpower-worlds-first-integrated-power-x-power-hydrogen-gas-turbine-demonstrator.html>

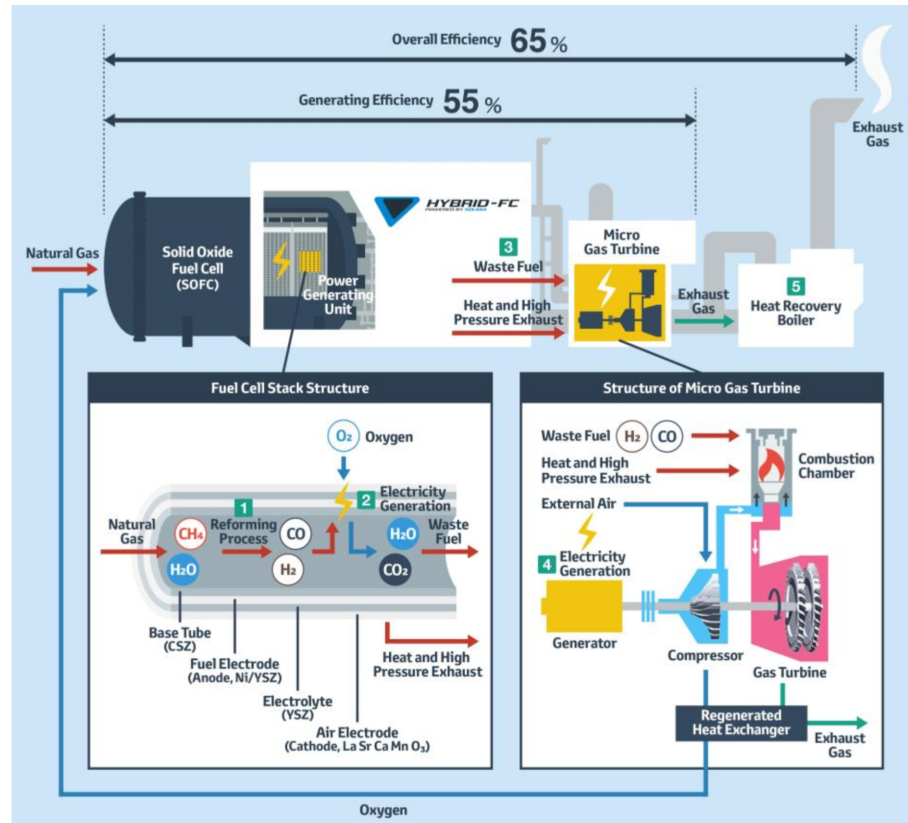
Hydrogen Use

❑ Gas Turbine × Fuel Cell

A synergistic integration of a high-temperature fuel cell (e.g., SOFC) with a micro gas turbine cycle

Toyota Starts Trial of a Hybrid Power Generation System Combining Fuel Cell Technology with Micro Gas Turbines at Motomachi Plant



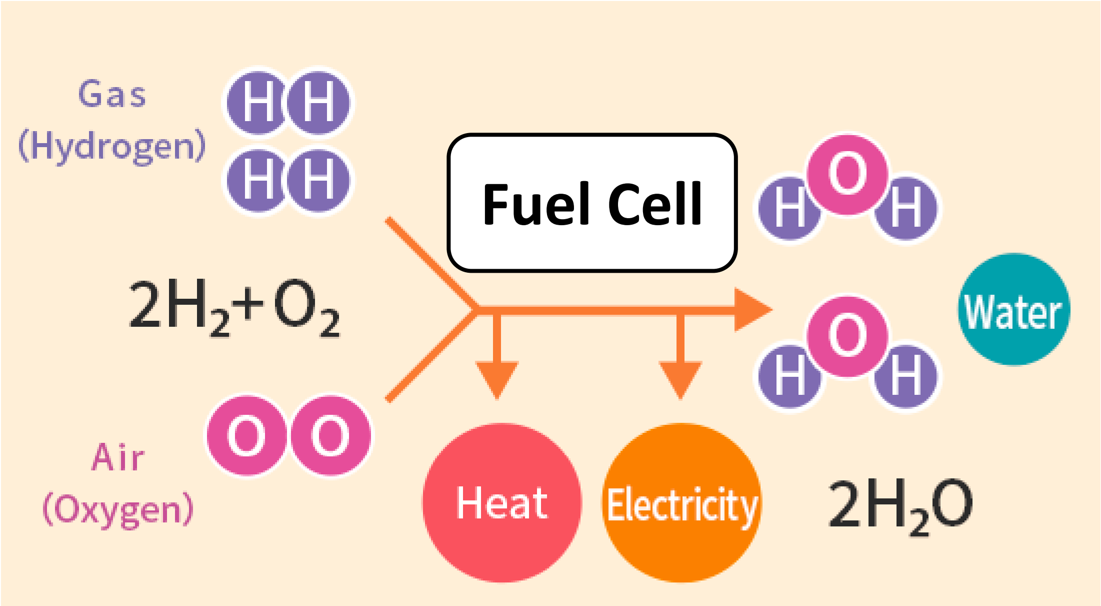


<Source: https://media.toyota.ca/en/releases/2017/toyota-starts-trial-of-a-hybrid-power-generation-system-combining-fuel-cell-technology-with-micro-gas-turbines-at-motomachi-plant.html>

Hydrogen Use

❑ Fuel Cell

An electrochemical energy conversion device that can directly converts chemical energy to electricity



– Water and/or CO_2 are produced from fuels such as H_2, C_mH_n, CO, NH_3, etc. with O_2 as the oxidant

– Partial conversion of the chemical energy difference between reactants and products into electricity

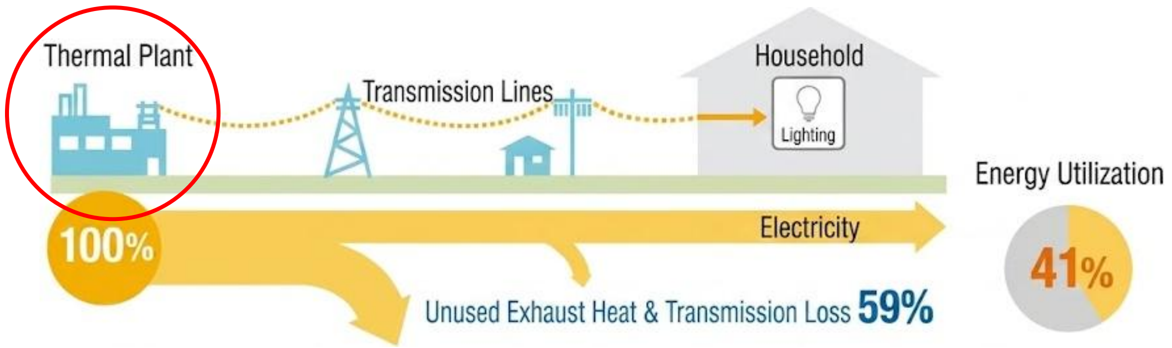
Fuel Cell

❑ Thermal Plant vs. Fuel Cell

An electrochemical energy conversion device that can directly converts chemical energy to electricity

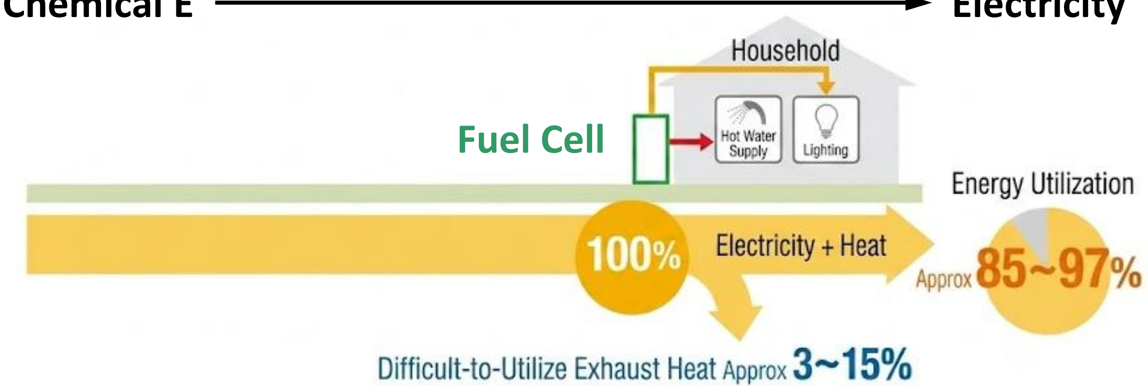
Combustion Turbine, Piston Generator

Chemical E → Mechanical E → Heat → Electricity



Electrochemical Reaction

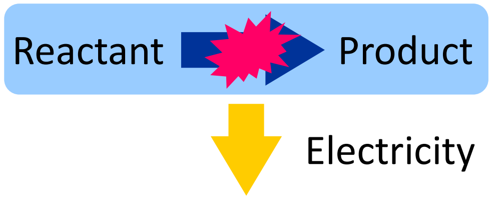
Chemical E → Electricity



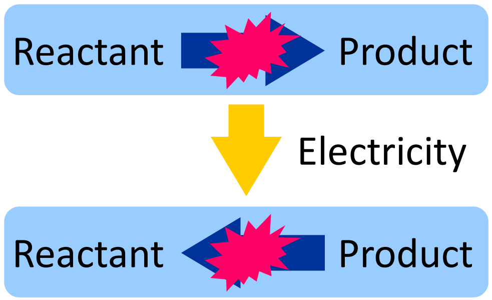
Fuel Cell

❑ Characteristics of Fuel Cell

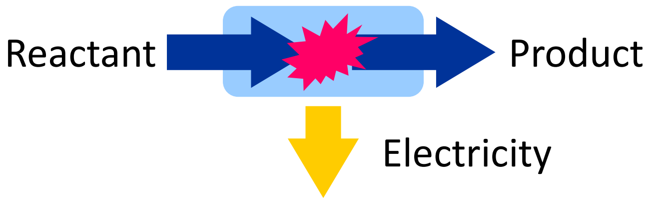
- **Primary Cell**
  Stores reactants and products inside the container
  Disposable (Single-use)
  (e.g.) Zinc–carbon (Manganese) battery, Alkaline battery



- **Secondary Cell**
  Stores reactants and products inside the container
  Reusable (Rechargeable battery)
  (e.g.) Li-ion battery, Lead–acid battery, Ni–Cd battery, etc.



- **Fuel Cell**
  Reactants are supplied from the outside
  Products are exhausted to the outside
  (e.g.) PEMFC, SOFC, etc.



Fuel Cell

❑ Characteristics of Fuel Cell

- **Continuous power generation is possible as long as reactants are supplied**
  Differs from conventional batteries (primary and secondary cells)

- **Expected Applications**
  Stationary power generation
  Vehicles / Mobile equipment (Motive power, Auxiliary power supply)
  Alternative to secondary batteries
  (Charging →Refueling: Saves time)
  (Higher fuel energy density leads to smaller and lighter systems)

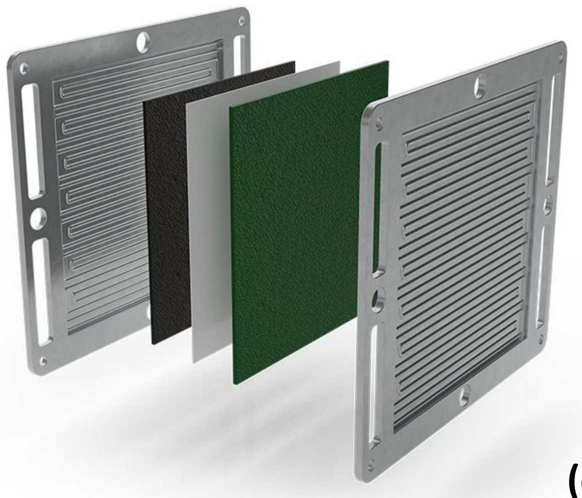
Fuel Cell

❑ Fuel Cell Applications

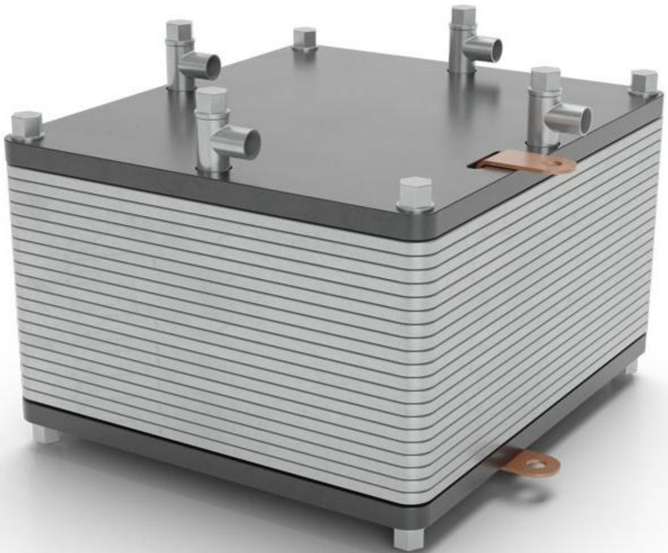
용량	고정 (발전소)	육상교통 (자동차, 열차)		수상교통 (선박)	공중교통 (비행기, 우주선)	
1~3 kW	가정용	전동기			드론	우주왕복선
20~100 kW	건물용	승용차		어선	0-2인[14]	우주정거장
200~400 kW[15]		상용차	동차/트램	여객선/잠수함	3-5인[16]	달 기지
1~4 MW	단지용		기관차	크루즈선	6-20인[17]	
4~10 MW					비행기	
40~100 MW				소형 화물선/군함/잠수함[18]		
100~1000 MW[19]	도시용			대형 화물선/군함/잠수함		

Fuel Cell

❑ **Single Cell & Stack**



Connected in series (e.g., 40-cell stack)



Nominal current: 30 A

Rated voltage: ~0.7–0.9 V

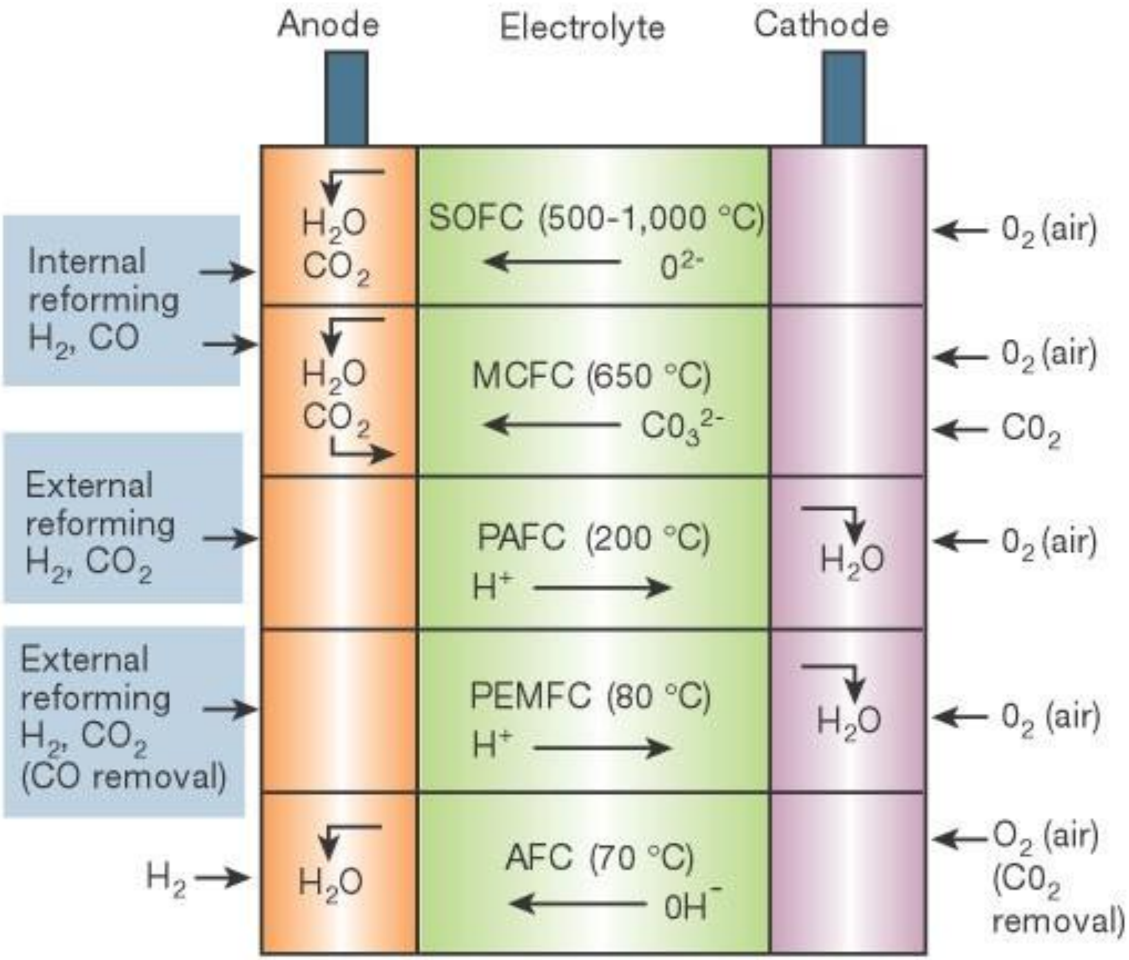
Rated power: ~25 W

Rated voltage: ~28–36 V

Rated power: ~1 kW

Fuel Cell

❑ Types of Fuel Cell



Nature, **414** (2001) 345–352

Fuel Cell

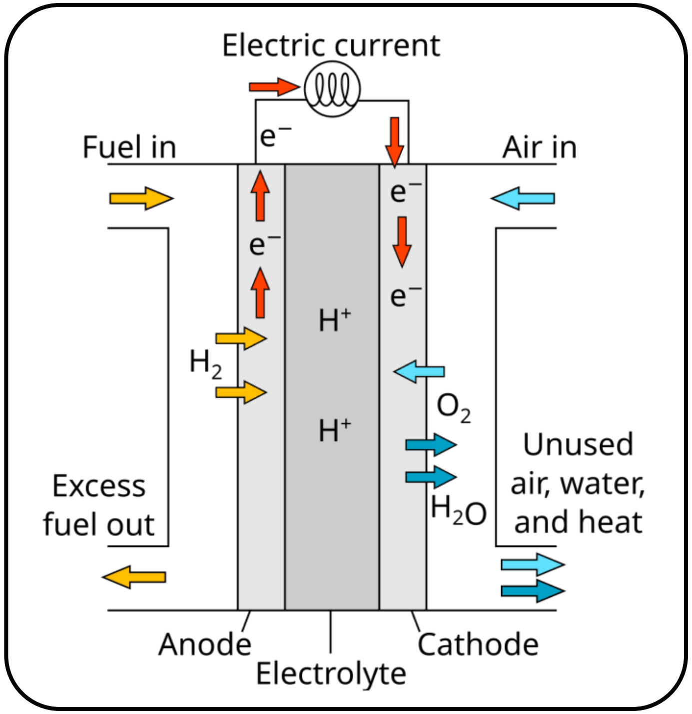
❑ Types of Fuel Cell

Type	Electrolyte	Stack Size	Eff (LHV)	Application	Advantages	Challenges
SOFC	**Stabilized zirconia**	**1 kW–2 MW**	**45–65%**	**Auxiliary power Electric utility Distributed generation**	**High eff Fuel flexibility Solid electrolyte Suitable for CHP Hybrid/GT cycle**	**High temp corrosion Rapid degradation Long startup time**
MCFC	Molten carbonate	300 kW–3 MW	45–60%	Electric utility Distributed generation	High eff Fuel flexibility Suitable for CHP Hybrid/GT cycle	High temp corrosion Rapid degradation Long startup time Low power density
PAFC	Phosphoric Acid	5–400 kW	35–42%	Distributed generation	Suitable for CHP Increased tolerance to fuel impurities	Expensive catalysts Long startup time Sulfur sensitivity
PEMFC	**PFSA (Nafion)**	**<1 kW–0.1 MW**	**30–40%**	**Backup power Portable power Distributed generation Transportation Specialty vehicles**	**Low temp Low corrosion Electrolyte mgmt. Quick startup & shutdown**	**Expensive catalysts Sensitive to fuel impurities**

Fuel Cell

❑ Types of Fuel Cell

- **H^+ Conducting Type (PEMFC)**

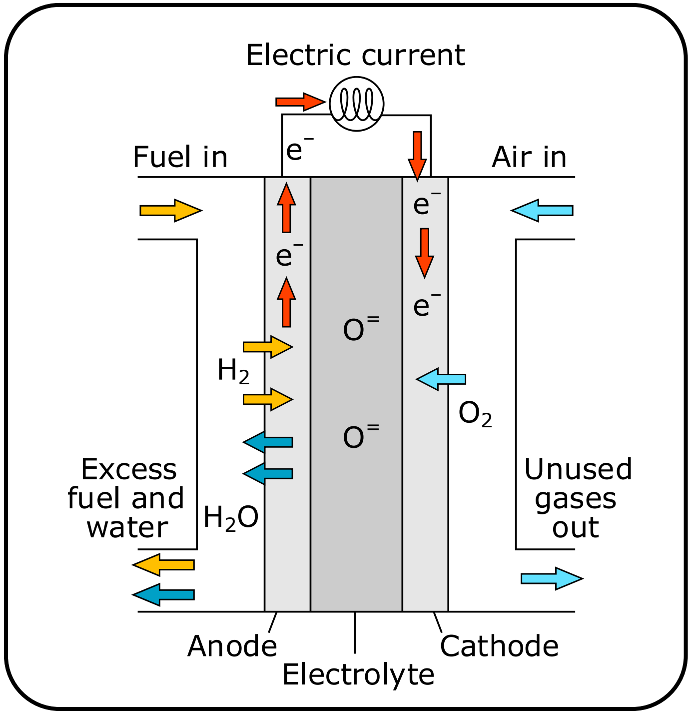


Anode: $H_2 \rightarrow 2H^+ + 2e^-$

Cathode: $0.5O_2 + 2H^+ + 2e^- \rightarrow H_2O$

Overall: $H_2 + 0.5O_2 \rightarrow H_2O$

- **O^{2-} Conducting Type (SOFC)**



Anode: $H_2 + O^{2-} \rightarrow H_2O + 2e^-$

Cathode: $0.5O_2 + 2e^- \rightarrow O^{2-}$

Overall: $H_2 + 0.5O_2 \rightarrow H_2O$

Fuel Cell

❑ The Very First Fuel Cell

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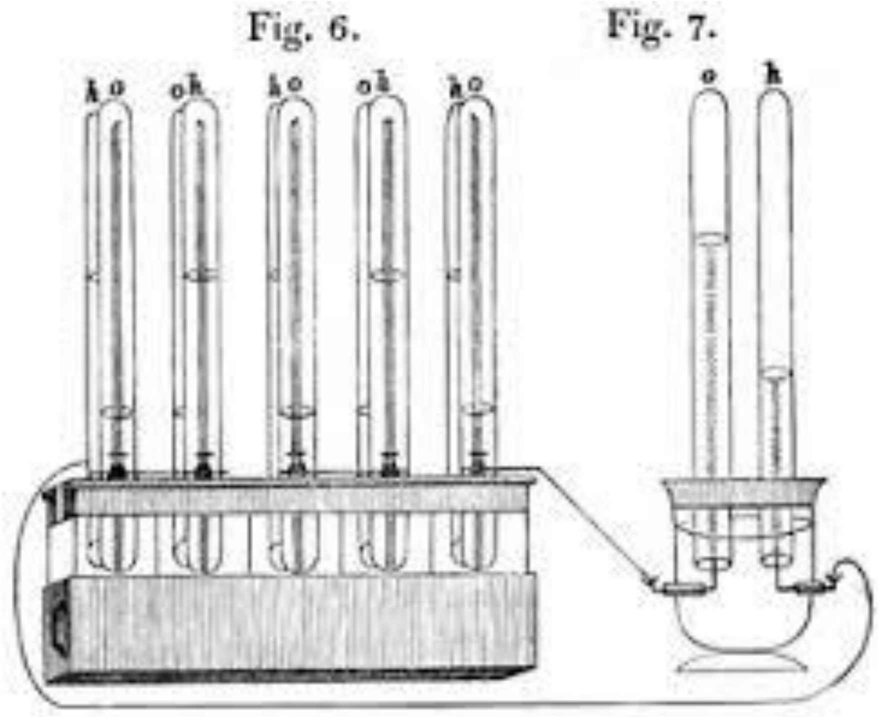
William Robert Grove
(1811–1896)



The Grove Cell (1838)
Used a Pt electrode immersed in HNO_3 and a Zn electrode in $ZnSO_4$ to generate about 12 A of current at about 1.8 V

<Source: http://americanhistory.si.edu/fuelcells/origins/orig1.htm>
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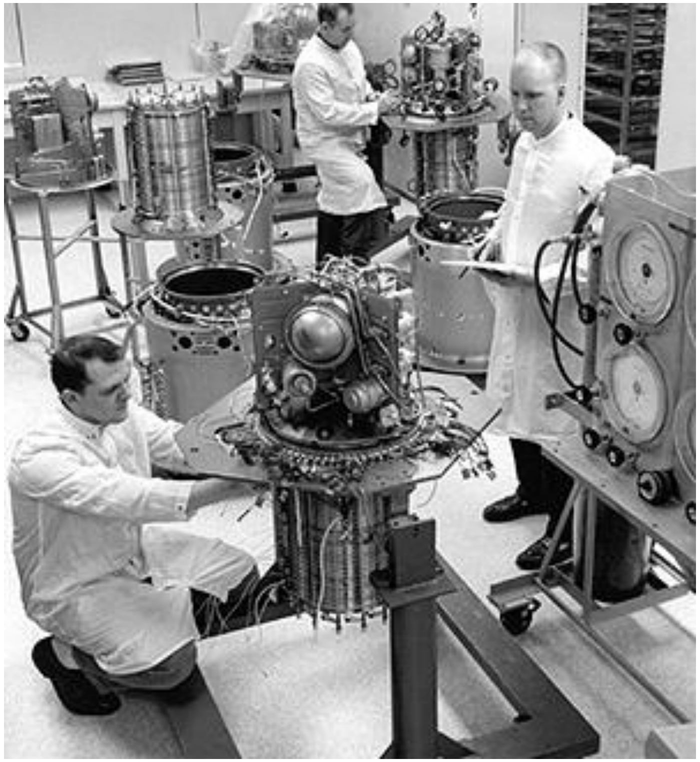
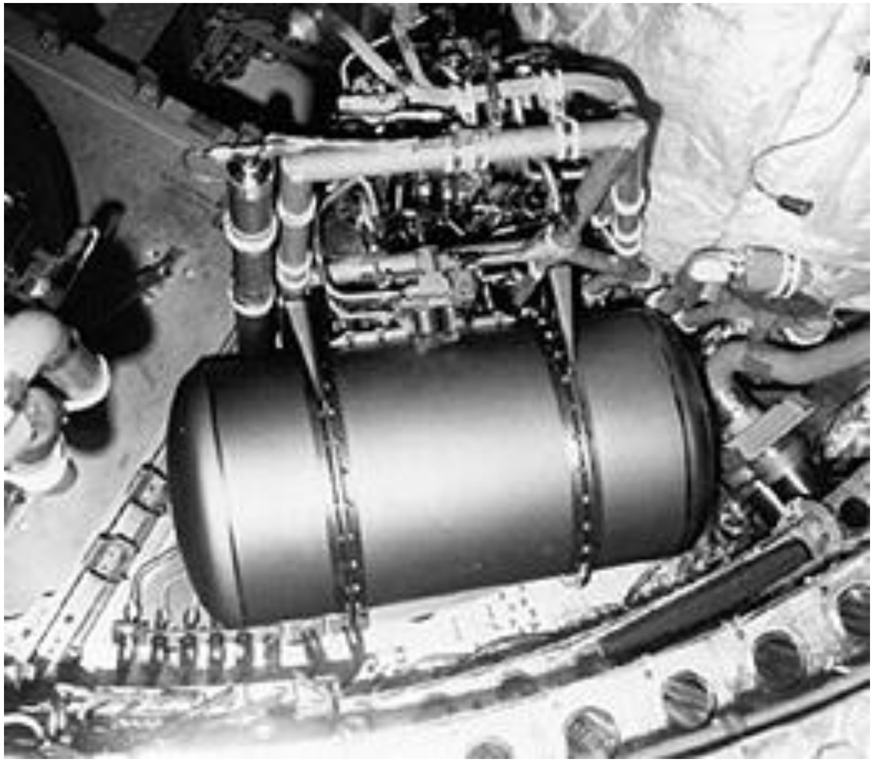
Fuel Cell

❑ First Applied in 1960s Space Exploration



Pratt & Whitney technicians assemble AFCs for Apollo service modules (1964)

<Source: https://americanhistory.si.edu/fuelcells/alk/alk3.htm >

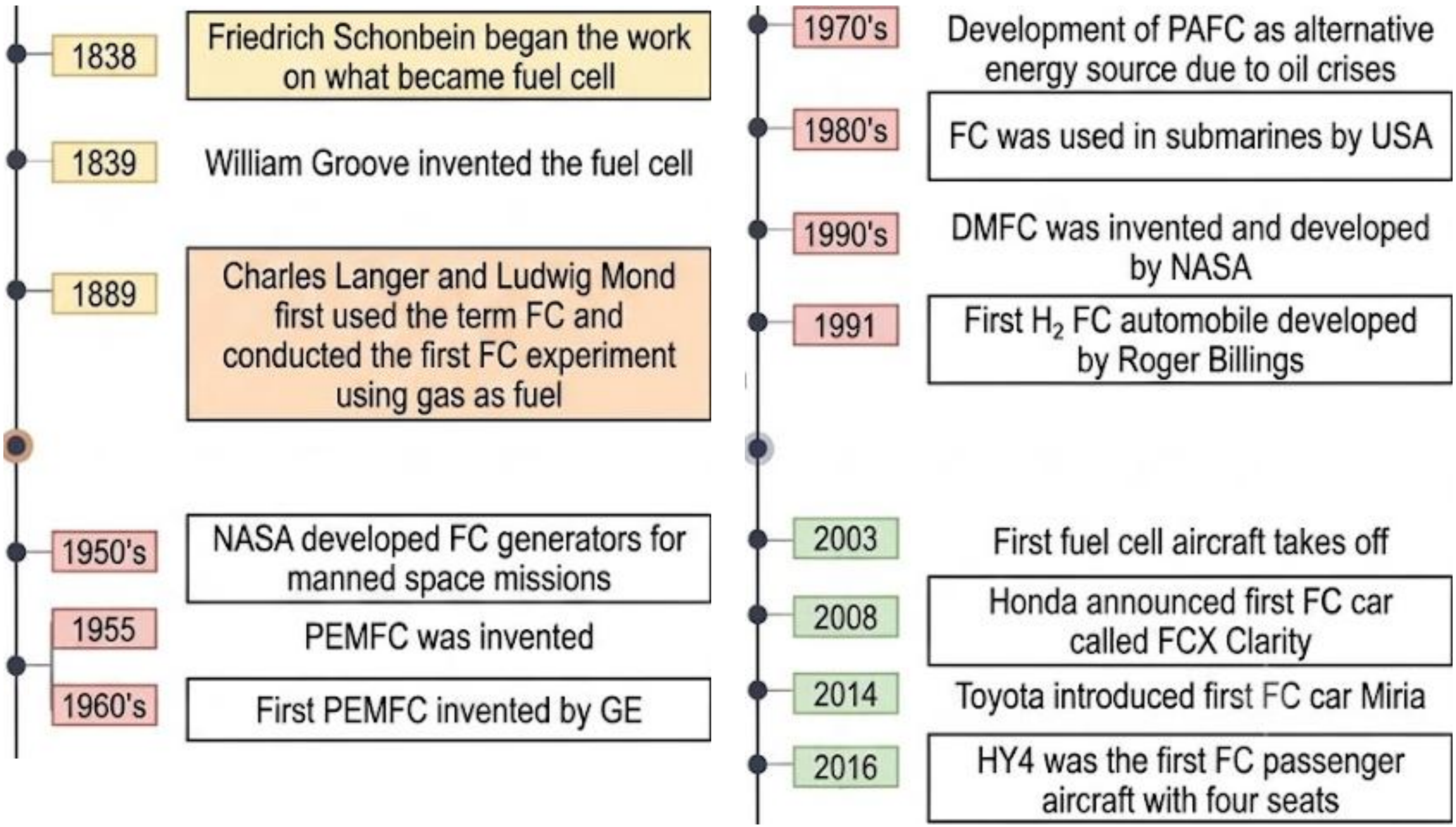


PEMFC in the Gemini 7 spacecraft (1965)

<Source: https://americanhistory.si.edu/fuelcells/pem/pem3.htm>

Fuel Cell

❑ Timeline



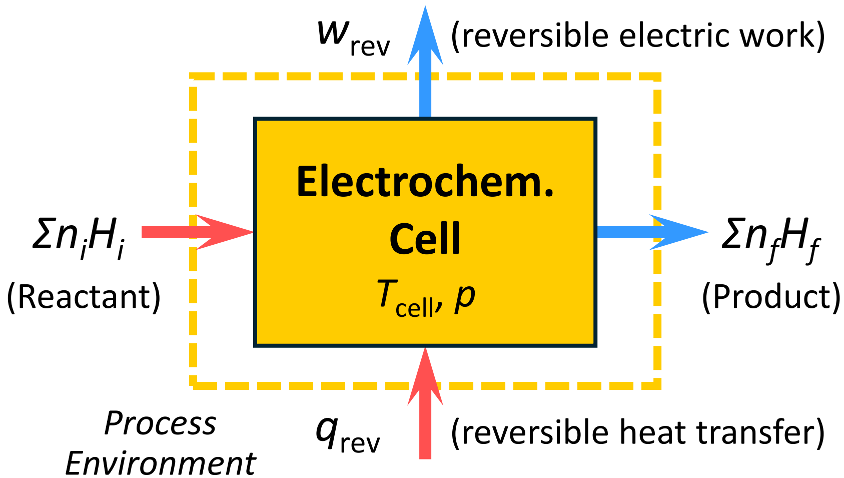
Int. J. Energy Res., **2024**(1) (2024) 7271748

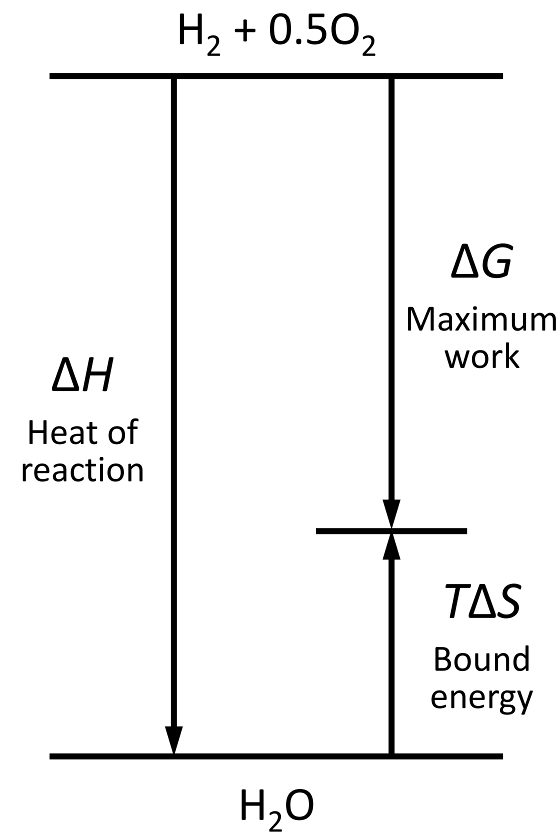
Fuel Cell

❑ Gibbs Free Energy & Cell Voltage

The system's thermodynamic driving force to perform electrical work is quantified by the voltage

- In an equilibrium system with an isothermal reversible reaction ($H_2 + 0.5O_2 \leftrightarrows H_2O$)





(1st law of TD) $q_{rev} - w_{rev} = \Sigma n_f H_f - \Sigma n_i H_i = \Delta H$

(2nd law of TD) $\Delta S = q_{rev} / T_{cell}$

(ΔH: enthalpy of the system
ΔS: entropy of the system)

Thus, the reversible work can be expressed by

$$w_{rev} = -\Delta H + T_{cell}\Delta S = -\Delta G$$

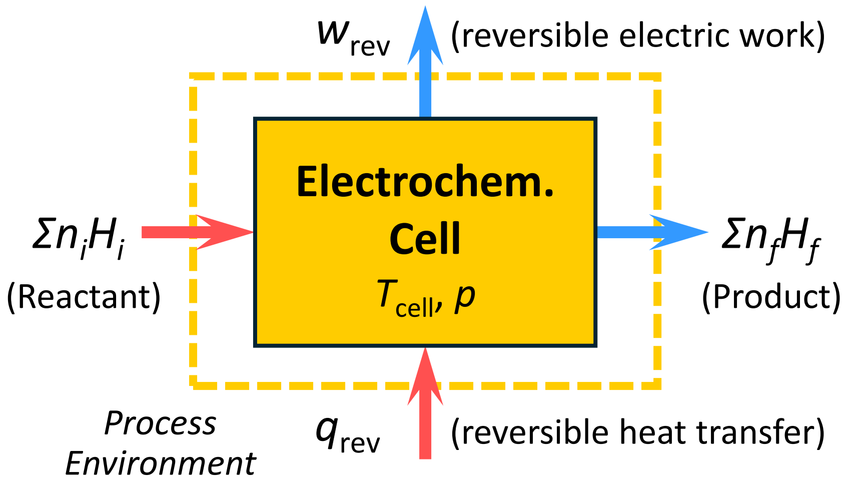
When $w_{rev} > 0 \Leftrightarrow \Delta G < 0$: Spontaneous (Fuel cell)
$w_{rev} < 0 \Leftrightarrow \Delta G > 0$: Nonspontaneous (Electrolyzer)

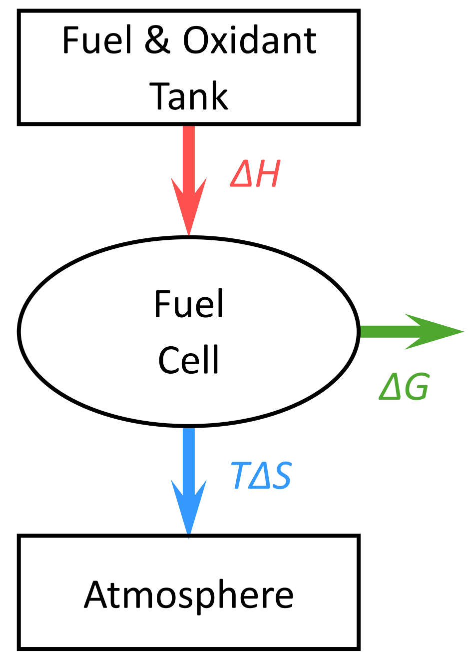
Fuel Cell

❑ Gibbs Free Energy & Cell Voltage

The system's thermodynamic driving force to perform electrical work is quantified by the voltage

- In an equilibrium system with an isothermal reversible reaction ($H_2 + 0.5O_2 \leftrightarrows H_2O$)





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When $w_{rev} > 0 \Leftrightarrow \Delta G < 0$: Spontaneous (Fuel cell)
$w_{rev} < 0 \Leftrightarrow \Delta G > 0$: Nonspontaneous (Electrolyzer)

$$\varepsilon_{\mathrm{FC,th}} = 1 - \frac{T\Delta S}{\Delta H}$$

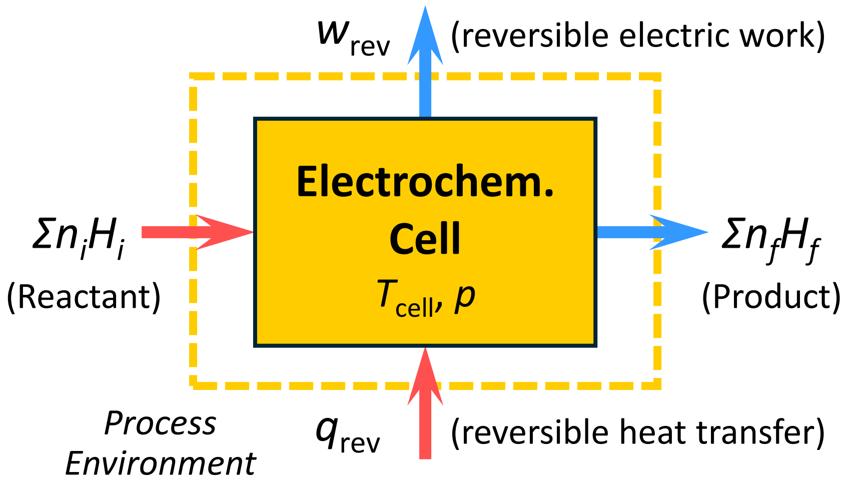
~80% @ 600 °C
~70% @1000 °C

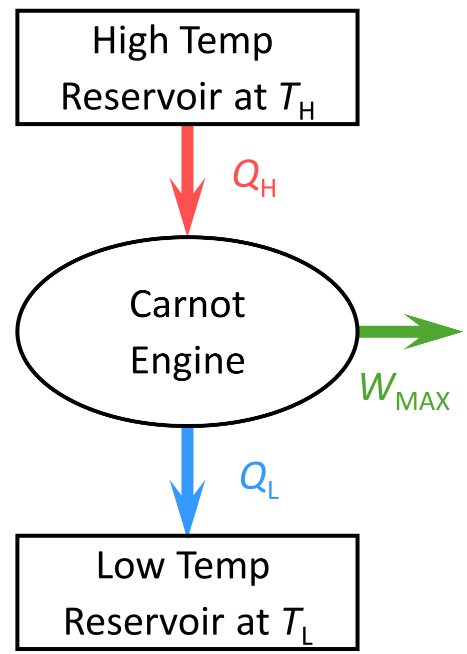
Fuel Cell

❑ Gibbs Free Energy & Cell Voltage

The system's thermodynamic driving force to perform electrical work is quantified by the voltage

- In an equilibrium system with an isothermal reversible reaction ($H_2 + 0.5O_2 \leftrightarrows H_2O$)





(1st law of TD) $q_{rev} - w_{rev} = \Sigma n_f H_f - \Sigma n_i H_i = \Delta H$

(2nd law of TD) $\Delta S = q_{rev} / T_{cell}$

(ΔH: enthalpy of the system
ΔS: entropy of the system)

Thus, the reversible work can be expressed by

$$w_{rev} = -\Delta H + T_{cell}\Delta S = -\Delta G$$

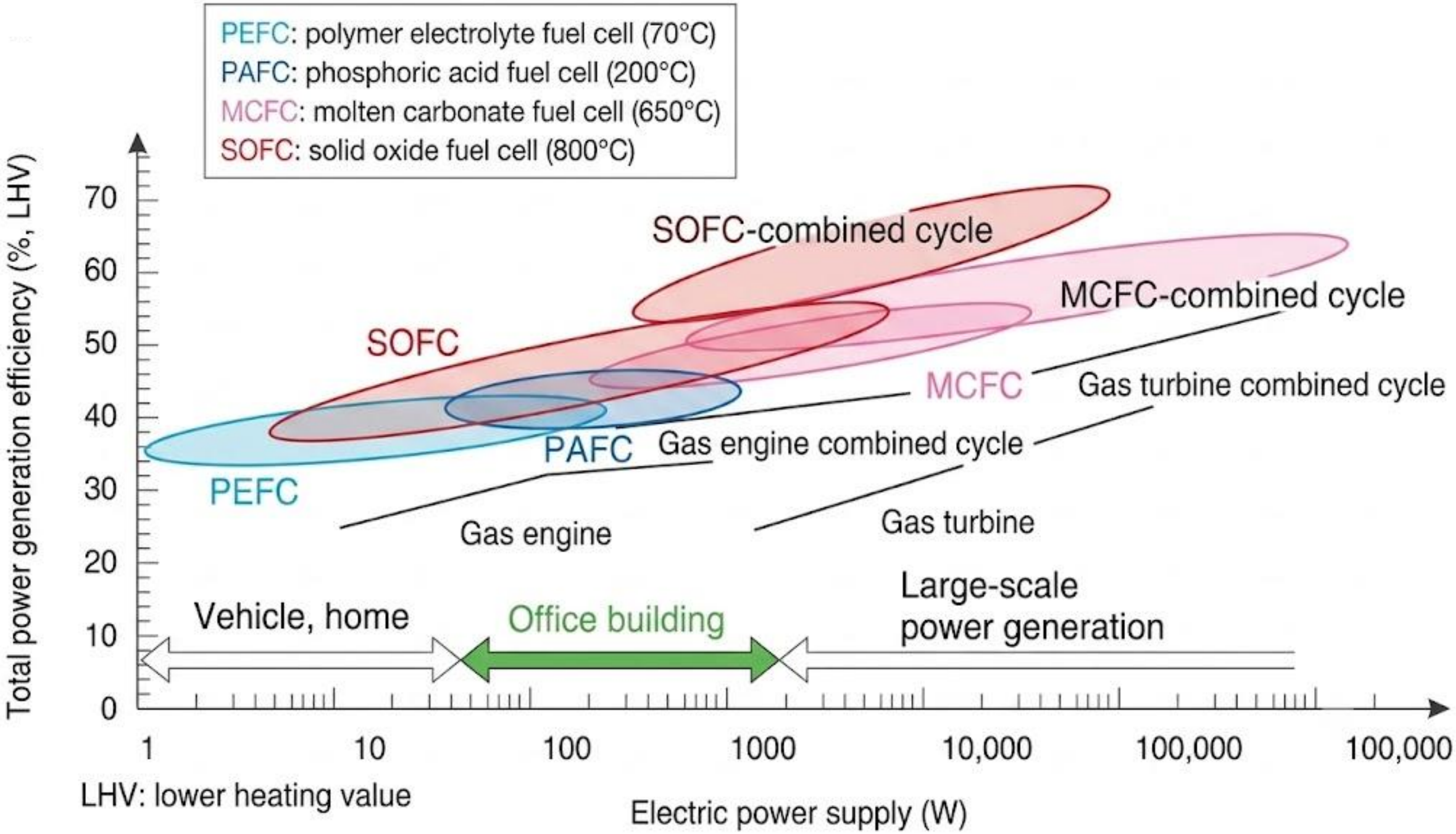
When $w_{rev} > 0 \Leftrightarrow \Delta G < 0$: Spontaneous (Fuel cell)
$w_{rev} < 0 \Leftrightarrow \Delta G > 0$: Nonspontaneous (Electrolyzer)

$$\varepsilon_{\mathrm{Carnot}} = 1 - \frac{Q_L}{Q_H}$$

~65% @ 600 °C
~77% @1000 °C

Fuel Cell

❑ Efficiency Comparison



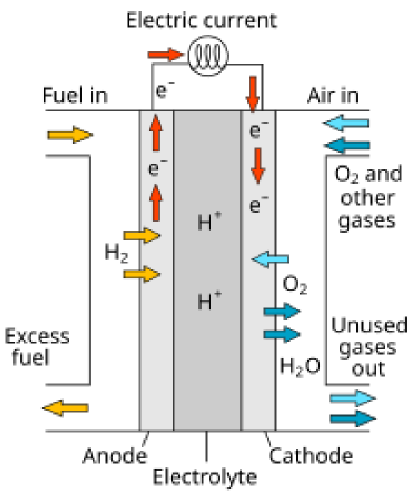
Int. J. Hydrogen Energy, **44**(58) (2019) 30591–30605

Fuel Cell

❑ Gibbs Free Energy & Cell Voltage

The system's thermodynamic driving force to perform electrical work is quantified by the voltage

- The electrical effect can be explained by thermodynamic principles



(Anode)
$H_2 \rightarrow 2H^+ + e^-$

(Cathode)
$0.5O_2 + 2H^+ + e^- \rightarrow H_2O$

(Overall)
$H_2 + 0.5O_2 \rightarrow H_2O$

The electric current I is expressed as,

$$I = \dot{q} = \dot{N}_{\mathrm{e}^-} e = (\dot{n}_{\mathrm{e}^-} N_{\mathrm{A}})e = 2\dot{n}_{\mathrm{H_2}} F$$

The electric power P_{rev} is the same as the thermodynamic quantity, so

$$P_{\mathrm{rev}} = V_{\mathrm{rev}} I = \dot{n}_{\mathrm{H_2}} w_{\mathrm{rev}} = \dot{n}_{\mathrm{H_2}}(-\Delta G)$$

Hence, the reversible voltage V_{rev} is

$$\therefore V_{\mathrm{rev}} = -\frac{\Delta G}{2F}$$

Under the assumption of ideal gas

$$\Delta G(T,p) = \Delta H(T) - T\Delta S(T,p),$$

where

$$S(T,p) = S^\circ + \int_{T_0}^{T} \frac{C_p(t)}{t}\,\mathrm{d}t - R\ln\left(\frac{p}{p_0}\right)$$

Then

$$\Delta G(T,p) = \Delta G^\circ + RT\ln\left(\frac{p_{\mathrm{H_2O}}}{p_{\mathrm{H_2}} p_{\mathrm{O_2}}^{0.5}}\right)$$

Thus, the electromotive force V_{rev} is

$$\therefore \boldsymbol{V_{\mathrm{rev}}(T,p) = -\frac{\Delta G^\circ(T)}{2F} - \frac{RT}{2F}\ln\left(\frac{p_{\mathrm{H_2O}}}{p_{\mathrm{H_2}} p_{\mathrm{O_2}}^{0.5}}\right)}$$

(e.g., $\Delta G°$=–237.14 kJ mol^{-1} ⇒ **$V_{\mathrm{rev}}°$=~1.23 V** at 25 °C, 1 atm)

Discussion

❑ EV vs. FCV

Discuss the differences between electric and fuel-cell vehicles